

Faster Optimal Power Flow Using Graph Neural Network-Assisted Methods

by

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DEDICATION

To my adoring son, my wonderful wife, and my loving family.

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I would like to express my heartfelt gratitude to Dr. Xingpeng Li for giving me the incredible opportunity to pursue my Ph.D. studies at the Renewable Power Grid Lab at the University of Houston. His guidance, advice, and unwavering encouragement over the past four years have been instrumental in shaping my research and my growth as a scholar.

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ABSTRACT

The electrical grid is transforming from a centralized system to a bidirectional network due to the rise of distributed energy resources like residential solar panels. The transformation, coupled with increased demand from electric vehicles and data centers, is putting significant strain on the aging electrical infrastructure. Insufficient investment in modernizing the grid, particularly transmission networks, further exacerbates issues related to reliability and security of the grid. Independent System Operators manage grid reliability using tools like Optimal Power Flow (OPF) to balance generation and demand efficiently. Increasing grid complexity and congestion makes solving OPF problems quickly a significant challenge. The complexity of the modern grid requires advanced computational methods to solve OPF problems quickly and efficiently.

Advanced methods are explored to optimize power system operations by reducing computational complexity in solving OPF problems. These problems are critical for determining generation dispatch points in real-time but become computationally intensive for large power networks due to the numerous variables and constraints involved. Proposed Graph Neural Networks (GNN), a machine learning approach, are utilized to reformulate the traditional OPF problems. The followings are key innovations:

- Congested Lines Prediction: GNN predicts heavily loaded or congested lines, allowing the optimization process to focus on critical lines constraints.

- *N*-1 Contingency Management: An augmented hierarchical GNN (AHGNN) is introduced to handle contingency scenarios, resulting in *N*-1 ROPF models that significantly cut computation time while preserving solution accuracy.
- Network Topology Optimization: By integrating GNNs with pre- and post-ML layers, the proposed approach accelerates network-reconfigured OPF (NR-OPF) solutions, improving both efficiency and accuracy.
- Maximum-Capacity Generators Prediction: A virtual node-splitting strategy is used to capture generator-level attributes, enabling a two-stage adaptive hierarchical GNN to further reduce constraints and variables, forming the ROPF Lines and Generators model.

Case studies confirm that these GNN-accelerated models outperform benchmark models in computational efficiency and solution quality, showcasing the potential of machine learning to enhance real-time grid operations.

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NOMENCLATURE

Sets:

G	Set of generators.
K	Set of lines.
N	Set of buses.
C	Set of contingency cases for $N-1$ OPF.
R	Set of congested lines for monitoring in $N-1$ OPF.
X	Set of congested lines for monitoring in ROPF.
Y	Set of generators with unknown generation level in ROPF.

Parameters:

P_g^{max}	Maximum generation limit of generator g .
P_g^{min}	Minimum generation limit of generator g .
P_y^{max}	Maximum generation limit of generator g . with unknown generation level in ROPF.
P_y^{min}	Minimum generation limit of generator g with unknown generation level in ROPF.
P_g^{ramp}	Ramping rate of generator unit g .
c_g	Cost of generation for generator unit g .
x_k	Reactance of line k .
$RateA_k$	Line rating limit of line k .

$RateC_k$	Line rating limit of line k in the contingency case of $N-1$ OPF.
$RateA_r$	Line rating limit of congested line r in the base case of $N-1$ OPF.
$RateC_r$	Line rating limit of congested line r in the contingency case of $N-1$ OPF.
$RateA_x$	Line rating limit of congested line x of ROPF.
d_n	Load profile at bus n .
M	Big-M, a very big number, for NR-OPF.

Variables:

P_g	Generation dispatch of generator g .
P_{cg}	Generation dispatch of generator g per contingency case c of $N-1$ OPF.
P_y	Generation dispatch of generator y in ROPF.
P_k	Power flow of line k .
P_{ck}	Power flow of line k per contingency case c of $N-1$ OPF.
P_r	Power flow of congested line r in the base case of $N-1$ OPF.
P_{cr}	Power flow of congested line r in the contingency case c of $N-1$ OPF.
P_x	Power flow of congested line x of ROPF.
$\theta_{f(k)}$	Phase angle of bus from line k .
$\theta_{t(k)}$	Phase angle of bus to line k .
$\theta_{f(ck)}$	Phase angle of bus from line k per contingency case c of $N-1$ OPF.
$\theta_{t(ck)}$	Phase angle of bus to line k per contingency case c of $N-1$ OPF.

$N1_{ck}$	Online status of each line k per contingency case c of N -1 OPF (binary).
NR_k	Switching status of each line k for NR-OPF (binary).

1. INTRODUCTIONS

1.1. Background

The power network is an integrated electrical infrastructure that encompasses generation, transmission, and distribution networks. Core elements of the network include generators, electrical loads, transmission lines, transformers, and reactive power compensation components such as capacitor banks. Due to the historical limitation of balancing between generation and load, electricity generation, delivery, and consumption must occur concurrently. This constraint poses substantial challenges for maintaining the reliability and operational security of traditional power systems in real time.

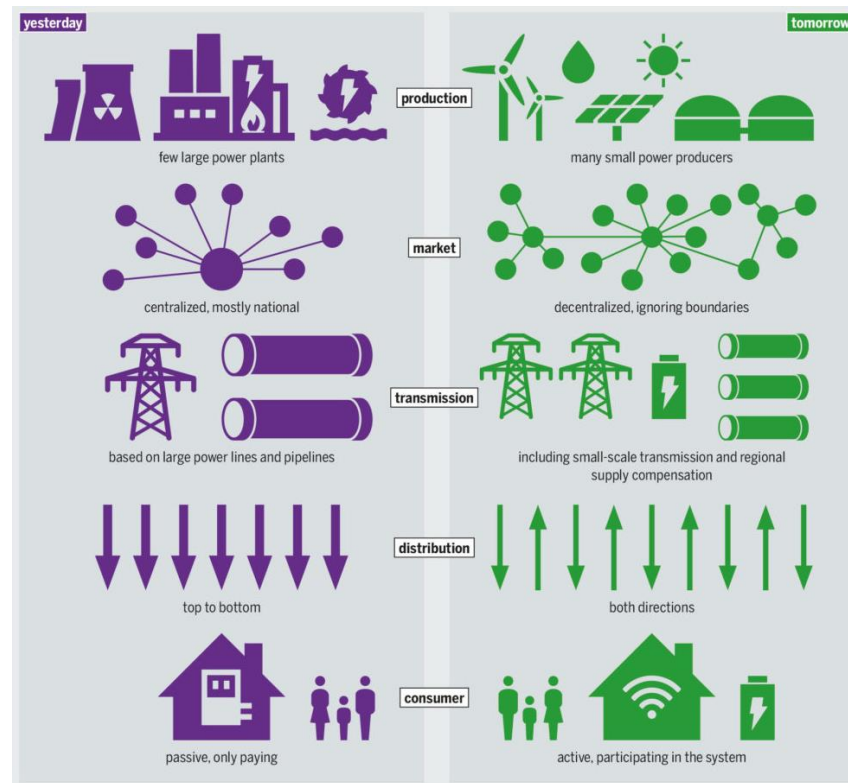


Fig. 1-1. Structural changes of the electrical grid [1].

The electrical grid is undergoing a significant transformation, transitioning away from the traditional centralized model. Historically, power flowed unidirectionally from large generators, through high-voltage transmission lines, and was stepped down for distribution to commercial and residential consumers. However, as illustrated in Fig. 1-1, the modern grid operates in a bidirectional manner. With the integration of distributed generation, residential consumers equipped with solar panels and battery storage systems can generate electricity during daylight hours, store excess energy, and utilize it at night. Surplus energy can also be fed back into the grid, particularly during periods of high demand, such as summer months.

The demand for electricity has surged in recent years, driven by two key factors. First, the electrification of the transport system has increased grid loads. Electric vehicles, while environmentally beneficial, impose significant power requirements for charging and operation. Second, the rapid expansion of artificial intelligence has triggered a construction boom in data centers, further straining electricity demand. Projections suggest that electricity demand could increase by nearly 10% by 2028 [2].

On the generation side, the renewable energy sector is experiencing unprecedented growth, fueled by tax incentives provided under the Inflation Reduction Act. Investments in renewable projects are essential for reducing greenhouse gas emissions and transitioning the grid to green energy sources. However, the intermittent and unpredictable nature of renewables introduces challenges in maintaining grid reliability. Large-scale battery storage systems only offer a partial solution to balancing load and generation during peak demand periods.

Despite substantial investments to meet increasing energy demands and integrate renewable generation, insufficient resources are being allocated to modernize the aging electrical infrastructure. Many components of the existing grid, such as the outdated 69 kV transmission network in Houston area, limit transmission capacity and exacerbate energy costs for consumers. Strengthening and upgrading these legacy systems is critical to ensuring a resilient, efficient, and future-ready electrical grid.

The summer of 2022 underscored the urgent need for transmission network upgrades within the ERCOT region. Extreme electricity demand during the hot summer months led to a sharp increase in energy prices across the four major metropolitan areas of East Texas. However, significant disparities in locational marginal pricing (LMP), the wholesale energy prices at specific locations, emerged between West Texas and East Texas. In West Texas, where abundant wind resources and numerous wind farms dominate, energy prices fell to near-zero. Conversely, in the Houston area, the cost per kWh surged to nearly four times the average. The volatility led to the bankruptcy of several electricity wholesalers who failed to accurately predict market conditions or implement viable pricing strategies.

A primary factor contributing to these LMP discrepancies is the limited number of high-capacity transmission lines connecting West Texas to Dallas and other urban centers in East Texas. Persistent congestion within ERCOT's transmission network restricts the flow of electricity from resource-rich areas to high-demand regions, diminishing grid efficiency [3]. In critical situations, the loss of a key transmission line or other congested pathways can exacerbate grid insecurity, potentially resulting in widespread blackouts across the ERCOT region. Due to the historic growth in demand

and frequent congestion, electricity costs are projected to rise by nearly 20% in the next 5 years [2]. These challenges highlight the pressing need for investment in transmission infrastructure to address lines congestion and enhance grid operational reliability.

1.2. Motivation

Independent System Operators (ISOs), such as ERCOT and CAISO, face increasing challenges in maintaining the balance between generation and load across the electrical grid. To ensure reliable and secure grid operation, ISOs employ various strategies to control and regulate the grid at different time scales. For efficiency, ISOs rely on real-time economic dispatch, a method that optimizes electricity generation based on economic cost to satisfy demand, while ignoring the transmission network.

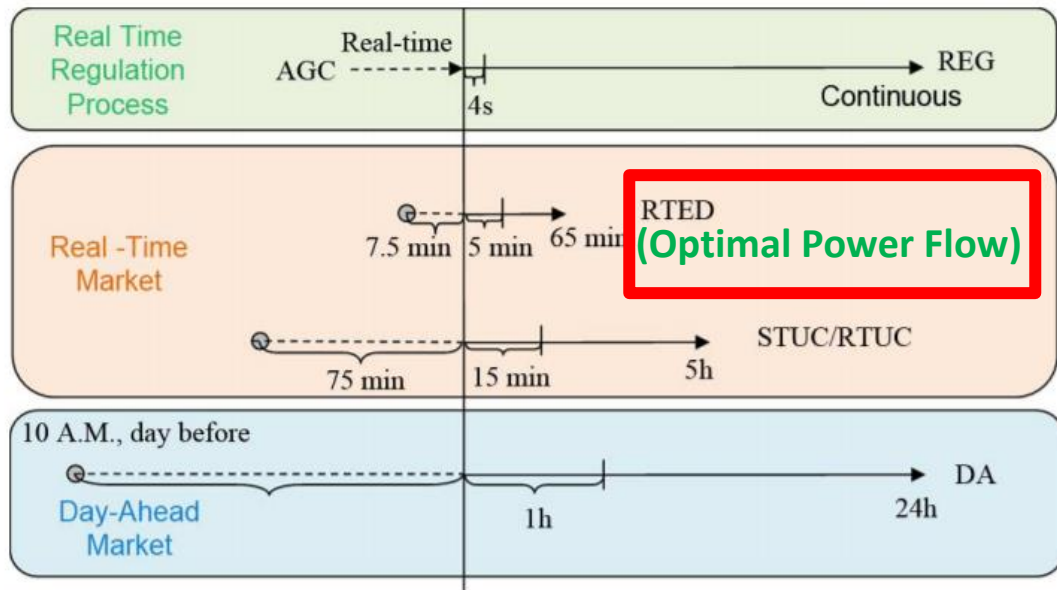


Fig. 1-2. Operating timelines of California Independent System Operator [4].

Based on in Fig. 1-2, the operational framework ISOs use for managing the real-time market is the Optimal Power Flow (OPF) method, which combines the power flow and economic dispatch problems. The formulation of OPF problems involves optimizing an objective function, commonly aimed at minimizing costs or losses, while satisfying a comprehensive set of operational and physical constraints, such as power balance, voltage limits, and thermal capacities of transmission lines. The versatility of OPF has led to its expanded applications across various domains of power system management. Modified OPF formulations, such as $N-1$ OPF and network-reconfigured OPF, are tailored to address specific operational challenges in power systems, further emphasizing the need for rapid computational approaches.

In $N-1$ OPF, operators must evaluate all possible single-component contingencies (e.g., the loss of a transmission line or generator) to identify scenarios that could lead to system unreliability or severe congestion. This requires computational methods capable of quickly assessing multiple contingency cases without compromising the accuracy of results.

For network-reconfigured OPF problems, the grid's topology is strategically modified to alleviate line congestion, address generation constraints, and optimize economic dispatch. This approach often involves dynamically rerouting power flows or adjusting network configurations to improve overall system efficiency and reliability. These specialized OPF problems highlight the need for advanced algorithms and computational techniques to support real-time grid operations in increasingly complex and dynamic networks.

The ability of OPF to address the complex interactions within modern power networks underscores its critical role in maintaining efficient and reliable grid operations. However, the increasing congestion and unpredictability of the modern grid present significant challenges. These factors complicate the ability to solve OPF problems quickly enough for real-time applications, highlighting the need for advancements in computational methods and system modeling to enhance grid responsiveness and reliability. The urgency of solving OPF problems efficiently has driven research into new methods to accelerate their resolution while maintaining solution optimality.

ML has become a well-established computational approach for addressing power system optimization challenges. ML models improve their predictive or decision-making capabilities by learning from extensive datasets without requiring explicit programming. These models are typically trained using sample datasets to identify patterns and generalize outcomes. However, traditional ML models often overlook the structural topology of the power network, which limits their applicability in grid-specific problems.

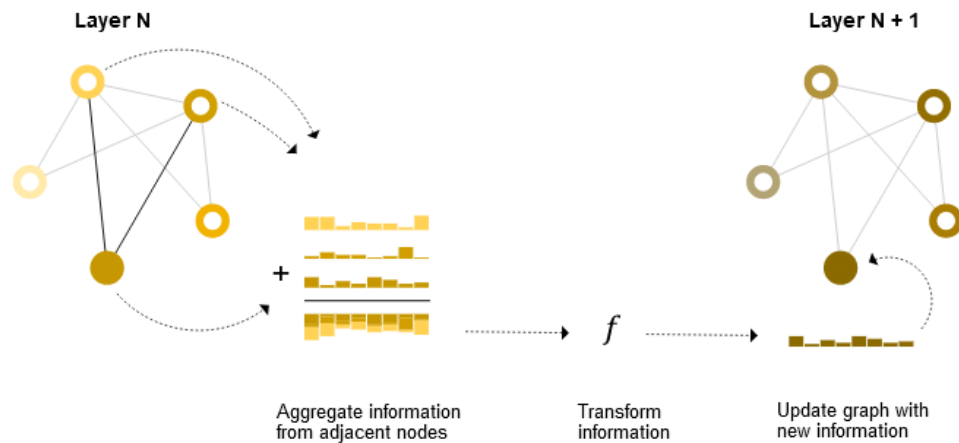


Fig. 1-3. Learning process of GNN through message embedding [5].

Graph Neural Networks (GNNs) represent a more advanced neural network architecture designed to handle network-structured data by leveraging graph topologies during training. From Fig. 1-3, GNNs utilize a process known as message passing, where information flows between nodes and edges of a graph across multiple layers of the neural network. For each node in the graph, GNNs aggregate embeddings messages from neighboring nodes and edges using an aggregation function. These aggregated messages are then processed through an update function to refine the node's state representation iteratively.

The integration of network topology into the training process allows GNNs to capture the spatial and structural relationships inherent in power networks. Various GNN architectures have been explored for power system control and optimization tasks, including OPF calculations. Studies have demonstrated that incorporating topology as an additional feature significantly improves the accuracy and efficiency of OPF predictions, highlighting GNNs as a powerful tool for addressing the challenges of modern power grids.

This dissertation extensively explores the applications of GNN models in power system optimization, with a particular focus on formulating and solving OPF problems. By leveraging the predictive capabilities of GNNs, constraints can be effectively eliminated, and the dimensionality of the problem reduced by lowering the number of variables. Various GNN architectures are applied and analyzed to enhance predictive accuracy and computational efficiency.

Additionally, the study investigates network topology reconfiguration to assess its effectiveness in improving the GNN model's performance. All solutions obtained

from the reduced OPF formulations are rigorously validated for optimality, ensuring no violation of operational or physical constraints. The reduced problem formulations demonstrate significant computational time savings in achieving optimal solutions, making them well-suited for real-time applications.

The dissertation also includes a detailed analysis and comparison of results across different GNN methods and configurations, verifying their correctness and demonstrating their potential for practical implementation in modern power systems.

1.3. Summary of content

The rest of this dissertation is structured as follows. Chapter 2 introduces a GNN approach to reduce the number of constraints in OPF problems, improving computational efficiency for large power systems. The GNN model predicts congested or heavily loaded lines based on load profiles and generation capacities, allowing only critical lines to be monitored and creating a reduced OPF (ROPF) problem. The proposed method significantly saves computation time while maintaining solution quality. A comprehensive analysis confirms that the GNN-enabled ROPF effectively reduces computing time without compromising the accuracy of the results.

In Chapter 3, it proposes an augmented hierarchical graph neural network (AHGNN) to predict congested lines for $N-1$ OPF, reducing computational time while maintaining solution quality in real-time grid operations. By creating an $N-1$ reduced OPF ($N-1$ ROPF), the AHGNN model efficiently handles large, complex power systems. Benchmark comparisons with other GNN-based ML models demonstrate the effectiveness of the AHGNN approach in optimizing performance. Case studies show

that the AHGNN-enabled $N-1$ ROPF significantly reduces computation time while preserving accuracy, highlighting the potential of ML, particularly GNN, in improving power system operations.

A GNN-based approach is used in Chapter 4 to accelerate solving network-reconfigured OPF (NR-OPF) problems, which traditionally require significant computational time due to their mixed-integer nature. The GNN model predicts optimal topologies offline, reducing complexity before optimization. An offline pre-ML filter layer enhances accuracy and reduces model size and training time, while a fast online post-ML selection layer refines predictions to select high-confidence solutions. Case studies demonstrate the proposed GNN-accelerated NR-OPF method, combined with pre-ML and post-ML layers, achieves superior efficiency and performance compared to traditional approaches.

Chapter 5 introduces a virtual node-splitting strategy to enhance GNN models by capturing generator-specific attributes, improving prediction accuracy for OPF problems. A two-stage adaptive hierarchical GNN predicts congested lines and maximum-capacity generators, significantly reducing constraints and variables in the ROPFLG model. Benchmark ROPF models, ROPFL and ROPFG, address reduced lines or generators independently. Case studies show ROPFLG outperforms these models and the full OPF, achieving substantial computational time savings while maintaining optimality and reliability in real-time grid operations.

The final Chapter 6 concludes the dissertation and discusses prospective research for future work.

2. REDUCED OPTIMAL POWER FLOW USING GRAPH NEURAL NETWORK

OPF problems are formulated and solved for power system operations, especially for determining generation dispatch points in real-time. For large and complex power system networks with large numbers of variables and constraints, finding the optimal solution for real-time OPF in a timely manner requires a massive amount of computing power. This paper presents a new method to reduce the number of constraints in the original OPF problem using a graph neural network (GNN). GNN is an innovative machine learning model that utilizes features from nodes, edges, and network topology to maximize its performance. In this paper, we proposed a GNN model to predict which lines would be heavily loaded or congested with given load profiles and generation capacities. Only these critical lines will be monitored in an OPF problem, creating a reduced OPF (ROPF) problem. Significant saving in computing time is expected from the proposed ROPF model. A comprehensive analysis of the GNN model's predictions was also made. It is concluded that the application of GNN for ROPF is able to reduce computing time while retaining solution quality.

2.1. Literature Review

Optimal power flow (OPF) is widely used for power system operations, analysis, and scheduling including real-time economic dispatch. OPF problems involve the optimization of an objective function that can take various forms while satisfying a set of operational and physical constraints. Applications of OPF have expanded because of its capabilities to deal with diverse aspects of the power system network. Due to the fast

increase in renewable energy and its intermittent nature, total available generation capacity can fluctuate widely throughout the day. OPF calculation must be done rapidly and accurately to meet given demands while satisfying the reliability of the system. Thus, improvements must be made upon the existing OPF model to meet the emerging challenging problems in power systems.

Different applications of machine learning (ML) in power system optimization have been studied extensively. For example, using recursive neural networks, researchers had tried to forecast energy prices of renewable energy sources for day-ahead energy market [6]; similarly, an integrated long-term recurrent convolutional network (LRCN) model was developed to forecast the wholesale electricity price [7]. The LRCN model is also utilized in [8] to estimate power system synchronous inertia. A multilayer neural network (NN) was proposed in [9] to predict power flow results as an alternative to traditional approaches. A logistic regression-based ML method is developed in [10] to determine a subset of generators' commitment status and thus reduce the day-ahead scheduling model. In [11], an NN-based battery degradation model is proposed for microgrid energy management.

Research in using ML for OPF has also been studied. In [12], an NN model was used to predict generations from load inputs in a security-constrained direct current OPF model. In [13], a convolutional neural network (CNN) model was built to predict generation dispatch using given load profiles. However, both methods do not account for topology of the power network. There is inherent information related to the global context of the network that can be used to train and enhance the ML model. Network

topology is crucial and correlates to how well a learning model performs under different conditions [14].

In this paper, we proposed an ML algorithm using graph neural network (GNN) to assist efficient formulation of the OPF problem. GNN is an advanced NN model that captures the dependence of graphs via message passing between the nodes of graphs. GNN model relies heavily on network topology to pass information from nearby nodes and edges across multiple GNN layers. In [15], a comprehensive overview of GNN in power systems was reviewed in detail for power system applications such as fault detection, time-series prediction, and power flow calculation. Evaluations to use different variants of GNN for research have also been proposed [16]. Recently, GNN has been used to predict wind speed for generations at wind farm cluster [17]. GNN has the potential to expand and improve upon existing tools for power system optimization such as OPF.

A recent attempt to use GNN for OPF calculation did not account for power flow and ignored the line rating limit in its formulation of the OPF problem [18]. For this paper, we designed a GNN model that can predict overloaded and congested lines in the power network with different given load profiles. Predictions from the GNN model are used to identify a subset of critical lines for supervision. We want to lower the number of monitoring lines or the number of line flow limit constraints in the OPF problem, in essence, producing a reduced OPF (ROPF) problem. With a ROPF model, it is expected that the amount of computing time for finding the optimal solution will decrease, especially for large and complex power systems. Model selection and performance maximization of the proposed GNN model are performed in this paper. A GNN model

was trained and tested against typical ML algorithms, such as NN and CNN, to assess and evaluate its overall accuracy.

The rest of the chapter is organized as follows. Background analysis of optimal power flow and GNN are covered in section 2.2. Section 2.3 introduces how sample data were generated, and the GNN model was developed. Section 2.4 provides analysis of the results and evaluates the performance of the proposed GNN model. Section 2.5 concludes the paper and describes possible future work.

2.2. Backgrounds

2.2.1. Optimal Power Flow

OPF model is often applied for real-time economic dispatch in practical power systems. It is used to optimize generation dispatch with given system loads while satisfying multiple constraints throughout the transmission network. The objective function in OPF usually focuses on minimizing operating costs. The OPF model below minimizes total system generation cost with variables of generation P_g and the associated cost c_g

$$\mathbf{Obj} : \min \sum_{g \in G} c_g P_g. \quad g \in G \quad (2-1)$$

The objective function is subject to multiple constraints. The constraint below

$$P_g^{\min} \leq P_g \leq P_g^{\max} \quad g \in G \quad (2-2)$$

sets the minimum/maximum generation limit of each generator in the system. The line flow on each line, P_k , is determined in

$$P_k = \frac{\theta_{f(k)} - \theta_{t(k)}}{x_k} \quad k \in K \quad (2-3)$$

and is constrained by the line limit rating, $RateA_k$, in

$$-RateA_k \leq P_k \leq RateA_k, \quad k \in K \quad (2-4)$$

$\theta_{f(k)}$ and $\theta_{t(k)}$ are the phase angles of the from-bus and to-bus of line k respectively. The nodal power balance denotes in

$$\sum_{g \in G(n)} P_g + \sum_{k \in K(n-)} P_k - \sum_{k \in K(n+)} P_k = d_n \quad n \in N \quad (2-5)$$

with the given nodal load, d_n .

For most OPF problems, the optimal solution is constrained by certain congested lines. These lines restrict the optimal solution within certain bounds and should be closely monitored. For most independent system operators, the constraint for line limit rating will only focus on a certain subset of lines that are heavily loaded or congested. These subsets of lines are chosen based on past solutions and past load profiles.

Our research focuses on how we can select these subsets of critical lines dynamically with respect to a given load profile using machine learning models, especially GNN. The GNN model should accurately predict which lines would be heavily loaded or congested given certain load profiles; then we only need to monitor those recognized critical lines rather than all lines in the OPF problem, which leads to an ROPF model relieving real-time computational burden. In summary, instead of using (2-4) to set line limit constraint for all lines, we will use

$$-RateA_x \leq P_x \leq RateA_x \quad x \in X \quad (2-6)$$

where x belongs to X , a subset of critical lines to monitor.

Using predictions from GNN model, we establish an ROPF model to determine the best combination of generation outputs to meet the given load with minimal cost

while maintaining line limit and nodal power balance constraints. The GNN model may be updated in near-real-time with different load profiles for different scenarios. It will allow system operators to achieve optimal generation dispatch within allocated computing time.

2.2.2. Graph Neural Network

ML is an established computer algorithm that has been researched and applied toward power system optimization [10]. ML models can automatically improve/learn through experience using a large amount of data. Models are trained using sample training data sets to make predictions or decisions without being explicitly programmed. CNN models and NN models are popular ML models that have been researched and used extensively for multiple applications across multiple disciplines. However, both models do not account for topology of the network in their predictions. For most electrical networks, topology contains crucial, inexplicit information/features that can be extracted/learned to enhance the learning model. GNN model has an advantage compared to the other two models due to its usage of adjacency matrix (nb by nb matrix, where nb denotes number of buses) to describe topology of the network. In addition, the global context of the network itself is used to develop relationship between multiple features of nodes and edges. In short, a GNN model is an optimizable transformation on all attributes of the graph (nodes, edges, global context) that preserves graph symmetries (permutation invariances).

In Fig. 2-1 below, the bus-branch model of a simple power system was transformed into graph format. Each bus is a node with information such as generation

and load represented as node features. Line flow and line thermal limit rating are edge features. The adjacency matrix characterizes the connection between each bus and its neighbors.

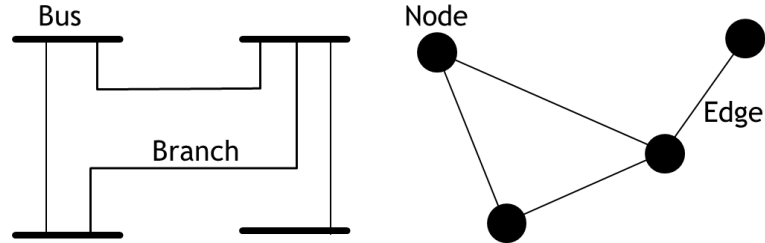


Fig. 2-1. Transformation of bus-branch model into graph format.

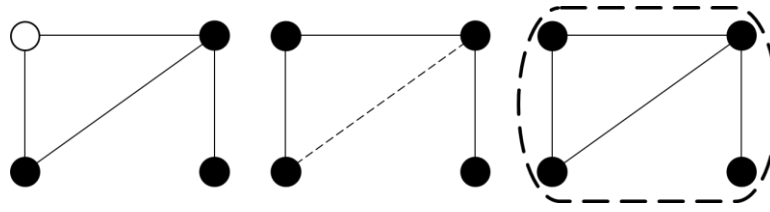


Fig. 2-2. Node-level vs. Edge-level vs. Graph-level classification task.

GNN has been used for multiple purposes. At the graph-level tasks, GNN is used to determine a single property for the entire graph. For example, GNN has been used extensively to predict the property of protein molecules based on their molecular structure, chemical bond, and polarity [19]. Similarly, for edge-level tasks, we want to identify the existence of an association between nodes. In citation networks, we explore the connections between research papers. We want to verify which of these nodes (research papers) share an edge (citation) or what the value of that edge is (number of citations).

2.3. Methodologies

To generate a large number of samples to sufficiently train and test a GNN model, we run normal OPF simulations on the IEEE 73-bus system with different load profiles to collect 20,000 samples. The load profile of each sample is varied within $\pm 10\%$ of the base load profile. From the 20,000 generated samples, the data set is divided into three groups: 80% for training, 10% for validation, and 10% for testing. Then, we used Pyomo library in python to solve for the optimal solution of each sample [20], [21]. Based on the solution, we created labels for each branch based on a pre-specified loading threshold that is defined as a percentage of the line rating limit. For example, if the line rating limit is 100MW and the loading threshold is 80%, a branch will be classified/labeled as heavily loaded/congested when its flow is over 80MW. The labels are represented as one-hot encoding and used as output during the training process.

For each sample, the node features are the nodal load, maximum and minimum generation at each bus, the number of branches (edges) connected to each bus, and bus types (load bus, generator bus and slack bus). The edge features include the line reactance and the line rating limit.

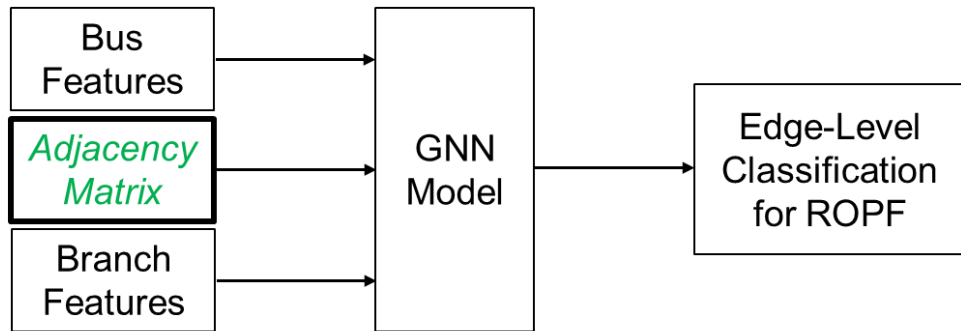


Fig. 2-3. Illustration of the GNN model. (4 XENet layers, 1 Dense layer).

As shown in Fig. 2-3, the proposed GNN model has four XENet layers followed by a dense layer. We decided to use XENet layer from the Spektral library as the primary layer to build the GNN model [22]. XENet layer is one of several popular GNN variants that have been developed recently for edge-level tasks. Using XENet, features from both nodes and edges are stacked and passed through each layer concurrently [23]. In addition, the adjacency matrix is passed along through each layer to retain the network topology and global context of the network during training. The final output is passed through a dense layer for edge-level classification. Once classification has been made, branches that are over the loading threshold will be included in the proposed ROPF model.

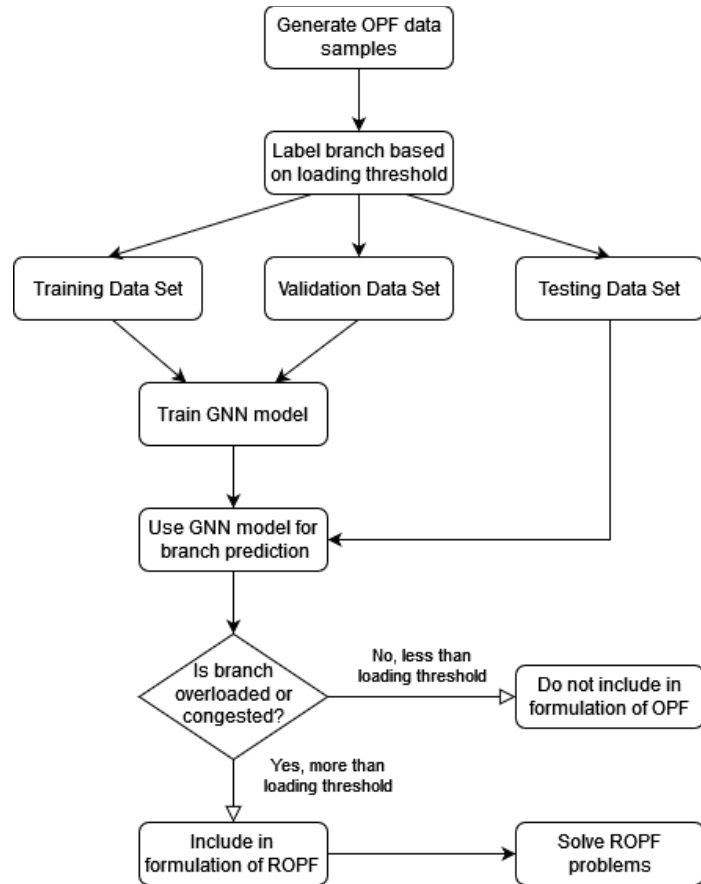


Fig. 2-4. Flow chart for constructing the proposed ROPF model using GNN.

2.4. Results and Analysis

In this section, we evaluated how well the GNN model performed when we set the threshold at 95% of line rating limit for classifying whether a line is heavily loaded or congested. Model training and OPF computation were done on a system with 4 cores i7-4790K CPU, GTX970 graphic card, and 32GB memory.

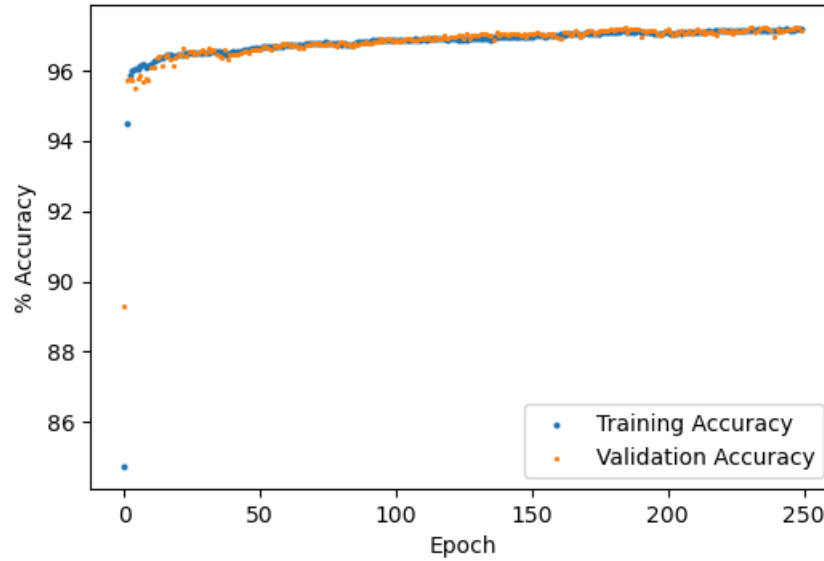


Fig. 2-7. Percent accuracy of prediction for training data set vs. validation data set during the training phase of GNN model.

From Fig. 2-7 above, we can see that the GNN model performs well during the training stage of 250 epochs. The training accuracy level is tracked consistently with the validation accuracy level.

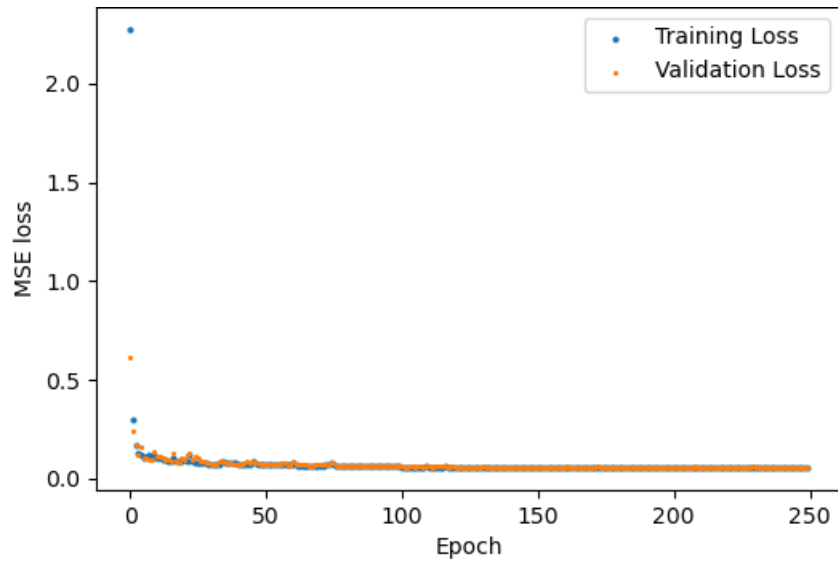


Fig. 2-8. MSE loss for training data set vs. validation data set during the training phase of GNN model.

As the training progress, the MSE loss dropped significantly within the first few epochs, then flattened and eventually saturated for the remaining epochs in Fig. 2-8.

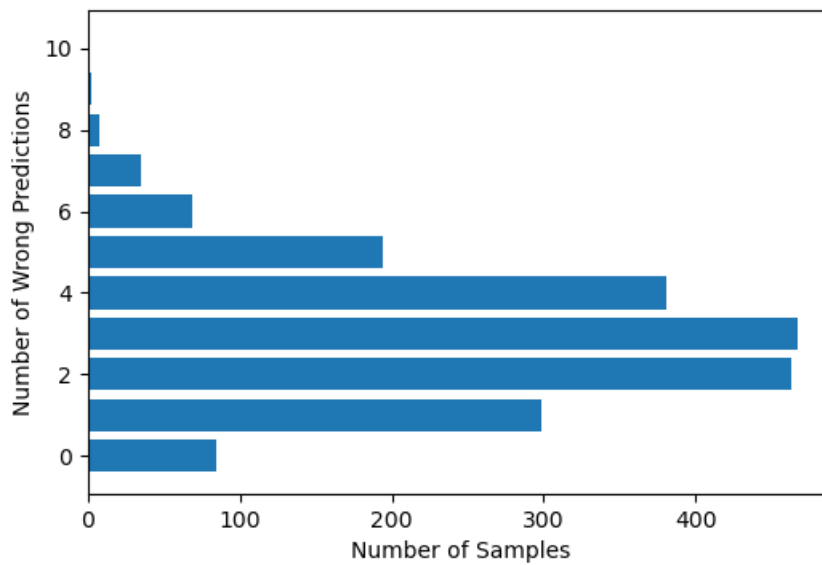


Fig. 2-9. Histogram represents the distribution of number of wrong predictions.

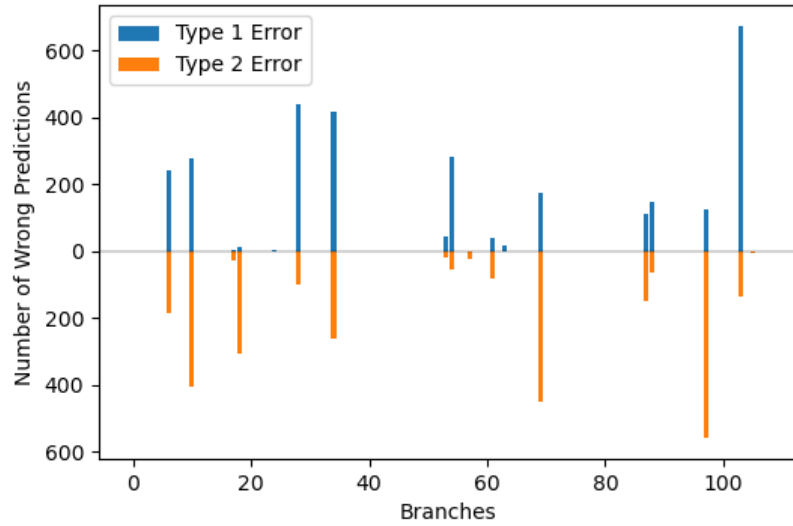


Fig. 2-10. Number of wrong predictions for Type 1 vs. Type 2 error per branch.

With a trained GNN model, it was used to make predictions against the testing data set of 2,000 samples. From Fig. 2-9 to Fig. 2-13, we analyzed predictions of GNN model with a loading threshold for branch classification set at 95%. For most of the samples in Fig. 2-11, there are wrong predictions on at least one branch to as many as eight branches.

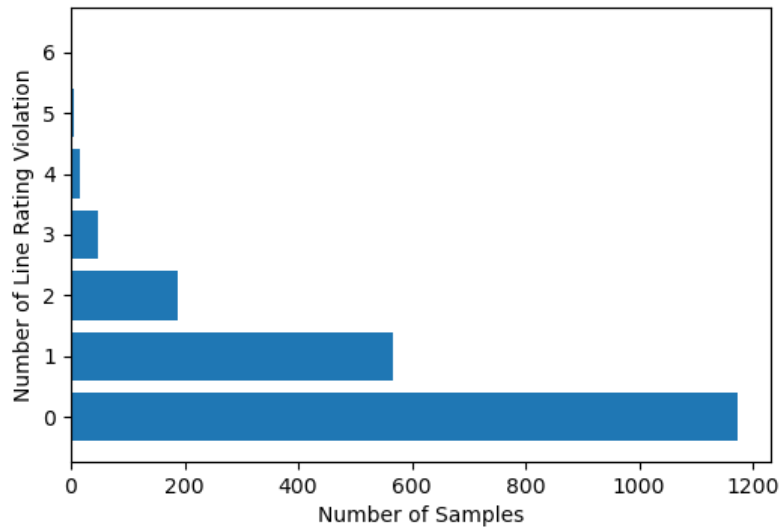


Fig. 2-11. The number of samples with at least 1 branch that violate line rating limit.

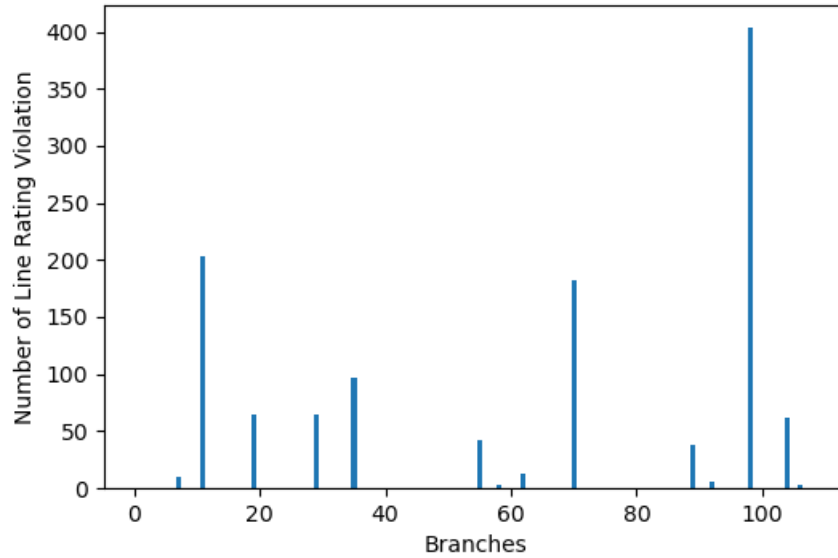


Fig. 2-12. Number of lines rating violation per branch.

For each branch of each sample, prediction is always made on whether the branch is congested or not. Thus, we would have two possible errors: (i) “false positive/type 1” error – incorrectly predicting a lightly loaded line to be heavily loaded, and (ii) “false negative/type 2” error – incorrectly predicting a heavily loaded line to be lightly loaded. From Fig. 2-10 above, we separated incorrect predictions into two separate types. Then, we went further by visualizing the total number of wrong predictions for each branch based on the type of error. We noted that most of the predicted errors occur in a small number of branches.

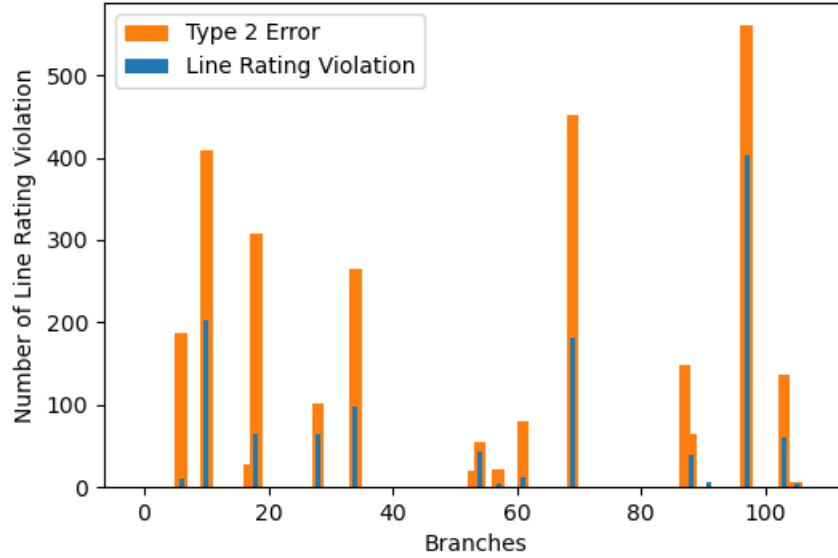


Fig. 2-13. Line rating violation is superimposed on top of type 2 error per branch.

Once predictions have been made, we ran ROPF calculations that only monitor branches that have been identified as congested by the GNN model. From the ROPF solutions in Fig. 2-11, it was noted that a significant number of samples contain unmonitored branches have violated the line rating limit. We then broke down the total number of line rating violations per branch in Fig. 2-12.

For type 2 error, we misclassified heavily loaded/congest lines. In turn, we did not include these lines in the group of monitored lines. Since these lines are critical, they are more likely to break the line limit constraint. In Fig. 2-13 above, Fig. 2-12 is superimposed on top of type 2 error in. We noticed a remarkable correlation, all branches that violate the line limit rating come from type 2 error almost exclusively

In Fig. 2-14, we shown the percent error prediction for different ML models at multiple thresholds. By setting multiple loading thresholds for branch classification, we wanted to determine which threshold would produce the lowest percent error in

prediction. Additionally, we wanted to test how well the GNN model performed compared to both NN and CNN model.

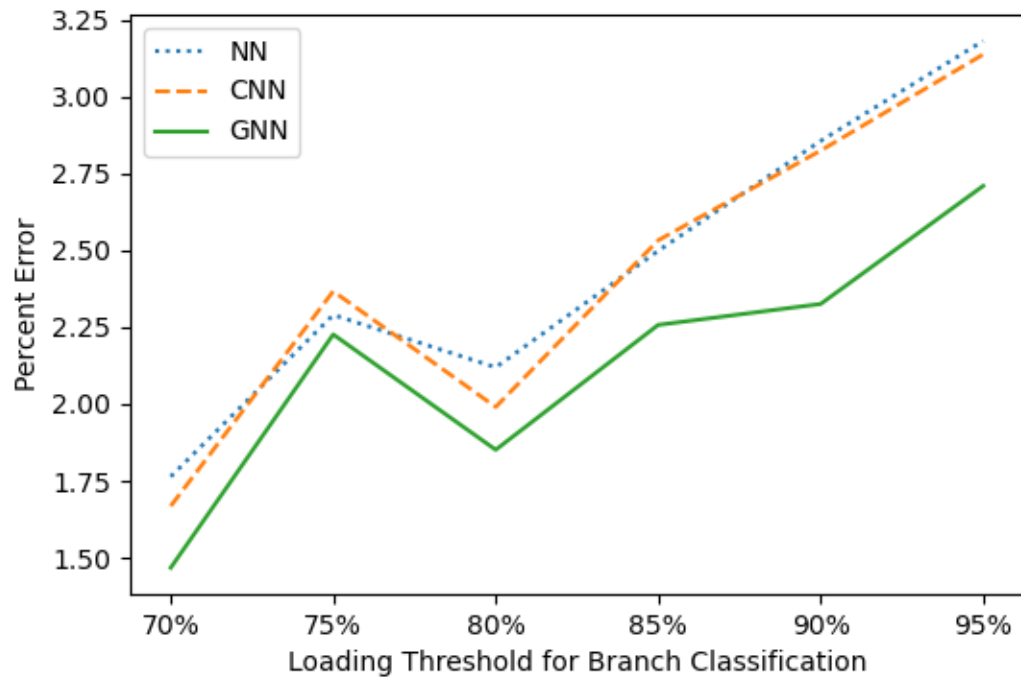


Fig. 2-14. Percent error of prediction at multiple loading thresholds for ML models.

TABLE 2-1 - Percentage prediction error for different models across multiple loading thresholds for branch classification.

	70%	75%	80%	85%	90%	95%
NN	1.76	2.29	2.12	2.50	2.86	3.18
CNN	1.67	2.37	1.99	2.53	2.82	3.14
GNN	1.47	2.23	1.85	2.26	2.33	2.71

Based on the data on TABLE 2-1, GNN model outperformed both NN and CNN model at every point of the loading threshold for branch classification. We also observed that there is a linear trend in which a higher loading threshold will likely lead to higher

prediction error overall across all ML models.

From Fig. 2-14 and TABLE 2-1, we can conclude that setting a high threshold for classifying congested lines will likely lead to a higher percentage of prediction error from the GNN model. With a high number of prediction errors, especially type 2 errors, there is a chance for an increase in the number of samples with unmonitored lines that violate the line limit rating.

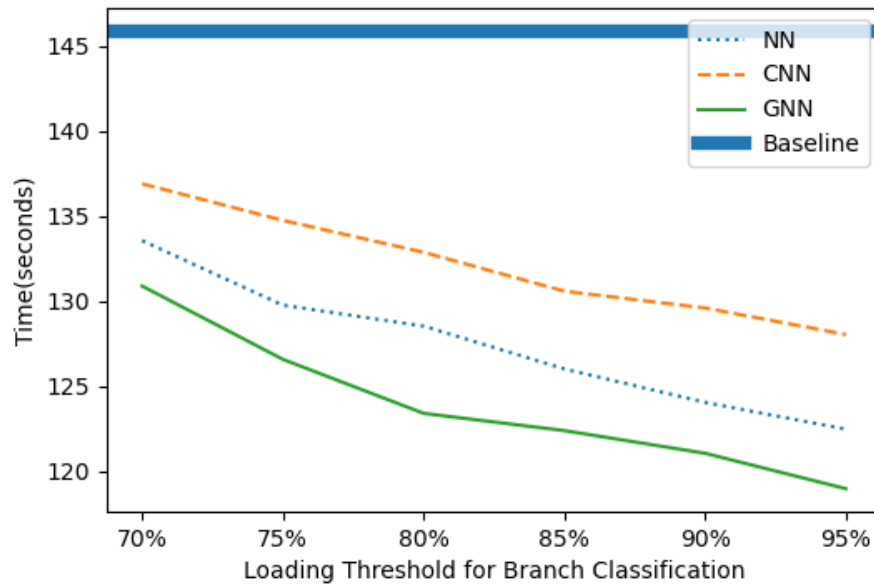


Fig. 2-15. Total time to solve 2000 ROPF testing samples at multiple loading threshold using predictions from different ML models.

However, there is benefit of choosing higher loading threshold for branch classification because it will reduce the number of critical lines in the subset of monitored lines. Looking at Fig. 2-15, we can clearly see there is a downward correlation trend between loading threshold and the amount of time it took to solve 2000 ROPF samples. ROPF performs better at every threshold compared to the baseline time of full OPF formulation problems.

TABLE 2-2 - Total time to solve 2000 ROPF samples at multiple loading thresholds using GNN prediction. Denote as a percentage of the full OPF problem.

Model		Time (s)	Time (%)
Full OPF		145.87	100.00%
ROPF	70%	130.88	89.72%
	75%	126.57	86.77%
	80%	123.39	84.59%
	85%	122.38	83.90%
	90%	121.05	82.99%
	95%	118.97	81.56%

From TABLE 2-2, at the highest threshold of 95%, it only takes 118.97 seconds to solve 2000 ROPF samples compared to the baseline time of 145.87 seconds for full OPF, a saving of almost 18% in computing time. However, when the 95% threshold was chosen in TABLE 2-3, it has the highest prediction error with only 12% of the total number of lines are monitored. We saw that almost 41% of the ROPF samples contain at least one branch that violates the line rating limit. These lines belong in the subset of unmonitored lines. We concluded previously that a high number of type 2 prediction errors is associated with a high number of ROPF samples with unmonitored lines that violate the line rating limit. Then, it is not recommended to set a 95% loading threshold for branch classification. Based on our observation, setting the threshold at 85% of line capacity produces the best result in terms of time saving and accuracy. At 85%, the computation time saving is 16%, a 2% reduction compared to 18% at 95%, but only

0.4% of the ROPF samples will contain at least one branch that violates the line rating limit. This is a significant gain in terms of accuracy with minimal loss in time saving.

TABLE 2-3 - Comparison of average measurement metrics of 2000 ROPF samples at multiple loading threshold using GNN model for branch classification.

Threshold	Time (%)	% of Samples Over Limit	% of Lines Monitored	Prediction Error (%)
70%	89.72	0.2	27.27	1.47
75%	86.77	0.2	23.66	2.23
80%	84.59	0.25	20.55	1.85
85%	83.90	0.4	17.67	2.26
90%	82.99	9.65	15.17	2.33
95%	81.56	41.35	11.99	2.71

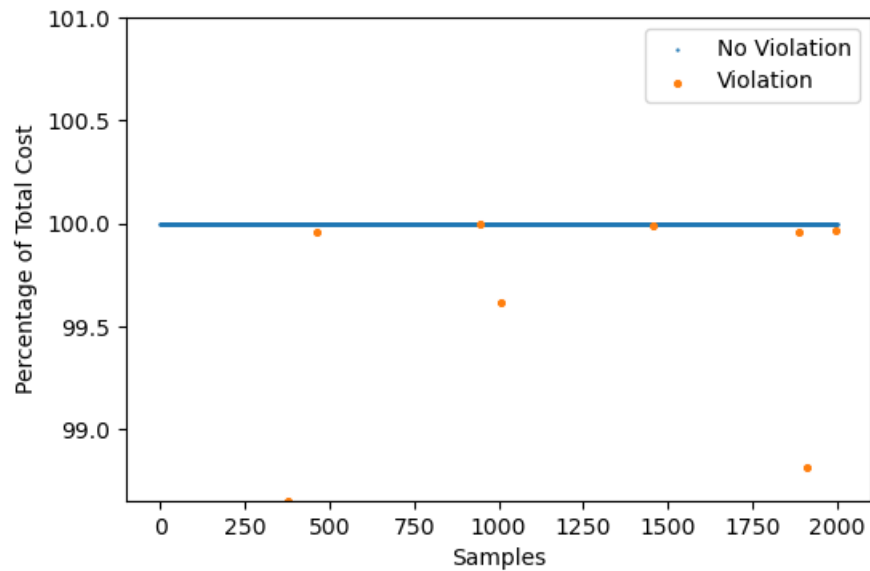


Fig. 2-16. Total generation cost of ROPF solutions at 85% loading threshold.

An interesting observation can be made regarding the figure below. At 85% loading threshold, only 0.4% or eight samples contain solution with at least one branch that violates line limit restriction. Fig. 2-16 depicts those eight samples. For these samples, a small reduction, less than 1%, in total generation cost is achieved at the cost of violating line limit restrictions.

2.5. Conclusions

From the error prediction result of all three ML models on the testing data set, we can conclude that GNN is the best model for classifying branches that will likely be overloaded or congested given a certain load profile. Having the lowest error prediction rate across all thresholds leads to additional saving in computing time of the proposed ROPF model. Choosing the right loading threshold for branch classification is another important step since accuracy matters as much as speed. At 85% threshold, predictions from the GNN model produce only 0.4% of solved ROPF cases that have at least one branch violating rating limit with a 16% reduction in computing time. This is a considerable improvement in terms of resources saving or computational cost.

In our research, we believe that model selection is an important step for the training of ML algorithms. GNN model, with its preservation of node features, edge features, and topology of the network during the training process, is a superior choice compared to NN and CNN. Building and training GNN models can be difficult, requiring large amounts of data, and computationally expensive, but the reward is significant. Prediction can be made quickly once the GNN model has been trained. Applications of GNN in power systems are still being explored and researched.

Regardless, the potentials of using GNN model to build and improve upon previous power systems technologies are substantial.

The proposed GNN model shows significant benefit in computing time saving for calculating OPF for real-time economic dispatch. With our experience in building and training GNN models, we believe that we can produce better results with higher computing time saving using a more complex and practical OPF model on much larger power systems. Due to the adaptability nature of GNN model, accurate predictions can still be made with different topologies of the power system network. We plan to develop a GNN model that quickly screens $N-1$ contingencies and detects lines that will violate line limit rating in the post-contingency situation.

3. $N-1$ OPTIMAL POWER FLOW USING AUGMENTED HIERARCHICAL GRAPH NEURAL NETWORK

Optimal power flow (OPF) is used to perform generation redispatch in power system real-time operations. $N-1$ OPF can ensure safe grid operations under diverse contingency scenarios. For large and intricate power networks with numerous variables and constraints, achieving an optimal solution for real-time $N-1$ OPF necessitates substantial computational resources. To mitigate this challenge, machine learning (ML) is introduced as an additional tool for predicting congested or heavily loaded lines dynamically. In this paper, an advanced ML model known as the augmented hierarchical graph neural network (AHGNN) was proposed to predict critical congested lines and create $N-1$ reduced OPF ($N-1$ ROPF). The proposed AHGNN-enabled $N-1$ ROPF can result in a remarkable reduction in computing time while retaining the solution quality. Several variations of GNN-based ML models are also implemented as benchmark to demonstrate effectiveness of the proposed AHGNN approach. Case studies prove the proposed AHGNN and the associated $N-1$ ROPF are highly effective in reducing computation time while preserving solution quality, highlighting the promising potential of ML, particularly GNN in enhancing power system operations.

3.1. Literature Review

Optimal power flow (OPF) constitutes a prevalent technique for power system operations, analysis, and scheduling. It encompasses the optimization of an objective function that can assume various forms while complying with a set of operational and physical constraints. Security constraints must be enforced in OPF to address power

system uncertainties, of which line contingency due to unforeseen circumstances is quite common in practice and may cause very severe outcomes. The applications of OPF have expanded due to its capacity to cope with diverse aspects of the power system network. $N-1$ OPF is used to address sudden major disturbances, particularly line outages, and ensure that the power grid is able to operate safely and serve all the demand in each possible post-contingency scenario [24], [25]. With $N-1$ OPF, power system reliability can be enhanced as compared to OPF with only reserve-based reliability requirement constraints.

Machine learning (ML) constitutes a field within artificial intelligence and computer science dedicated to harnessing data and algorithms to simulate human learning processes, with the goal of progressively enhancing its accuracy over time. The adoption of ML technologies in the power system domain has been growing very fast in recent years. For example, long-term recurrent convolutional network (LRCN) models have been developed for electricity price prediction [6], [7], load forecasting [26], and synchronous inertia estimation [8]. Deep neural network (DNN) and graph neural networks (GNN) have been proposed to predict power flow results, offering an alternative to traditional methods [9]. Logistic regression and multiple deep learning methods have been employed to determine the commitment status of a subset of generators with confidence, which can accelerate the computing process of solving day-ahead scheduling models that select the most cost-effective generation units [27], [28].

There are some prior efforts that study the use of ML for OPF in the literature. In [12], a DNN model was used to predict generations from load inputs in a security-constrained OPF model. In [13], a convolutional neural network (CNN) model was built

to predict generation dispatch given load profiles. However, both methods do not account for the topology of the power network. There is inherent information related to the global context of the network that can be used to train and enhance the ML model. Network topology is crucial and correlates to how well a learning model performs under different conditions [14].

GNN that was designed to address network-structured data can utilize network topology as part of the features during the training process [29]. GNN is an advanced neural network model that captures the dependence of graphs via message passing between the nodes of graphs. The GNN model relies heavily on network topology to pass information from nearby nodes and edges across multiple GNN layers. In [15], it presents a comprehensive overview of GNN for power system applications such as fault detection, time-series prediction, and power flow calculation. Different variants of GNN have been evaluated in [16] for possible applications in power system control and optimization. As an example, one variant of GNN has been used as a prediction tool for the power flow model [30]. Recently, another variant of GNN has been used to predict wind speed for generation at wind farm clusters [17]. GNN has the potential to expand and improve upon existing tools for power system optimization such as OPF.

In both [31] and [32], network topology is a key factor in calculating OPF using ML. It shows that utilizing the topology as an additional feature will enhance prediction for OPF calculation. A recent attempt to use GNN for OPF calculation did not account for power flow and ignored the line rating limit in its OPF formulation [18]. There were attempts to compute AC OPF using an unsupervised GNN model or deep reinforcement learning [33], [34]; however, constraint errors were still present in their solutions.

In our previous work [35], we proposed an ML algorithm using GNN to assist efficient OPF formulation. Using GNN, we created a reduced OPF (ROPF) model that decreases computing time by 16% compared to full OPF, while the ROPF solutions satisfy all the constraints.

In this paper, we proposed to use GNN for the $N-1$ OPF problem, which evaluates possible $N-1$ contingency scenarios along with the base scenario where no contingency occurs. For the $N-1$ OPF model, the numbers of constraints and variables increase exponentially and are significantly higher compared to the base-case OPF, which requires advanced methods to efficiently solve $N-1$ OPF. Particle swarm optimization [36] and compensation method [37] are two notable methods to simplify the $N-1$ OPF model. However, the particle swarm optimization method did not consider computing time as one of its evaluation metrics while the compensation method did not consider all possible contingencies or objective total cost in its result.

To implement the $N-1$ OPF algorithm using predictions from ML, a sophisticated and highly complex GNN model is needed to predict congested lines for the base case and the $N-1$ contingency cases. In this paper, we proposed an augmented hierarchical graph neural network (AHGNN) to effectively identify and eliminate unnecessary non-binding constraints for $N-1$ OPF. Previous applications of hierarchical models in deep learning, such as classifications of bioimaging data, produced significant improvement in accuracy compared to a straightforward flat ML model [38], [39]. A hierarchical deep learning model was also recently used for battery degradation predictions, which demonstrated better inference performance than single-stage ML models [40].

The proposed AHGNN model is a nested two-stage hierarchical ML model. The first-tiered GNN model, base-case GNN (BC-GNN), is trained to predict congested lines in the normal pre-contingency base case. These predictions from this first BC-GNN model serve a dual purpose: (i) identify a subset of critical lines for further monitoring and (ii) are utilized as crucial features for the second-tiered GNN model that is a contingency-case GNN (CC-GNN) model. This CC-GNN model is employed to predict congested lines for each possible $N-1$ contingency case. These predictions collectively aim to reduce the number of monitoring lines or constraints related to line flow limits in the $N-1$ contingency cases. Using these predictions, the full $N-1$ OPF problem is effectively transformed into an $N-1$ reduced OPF, or $N-1$ ROPF problem. Using the trained GNN models on samples on the validation data set, predictions were made on the base cases and the $N-1$ contingency cases. Subsequently, those predictions were used to create $N-1$ ROPF. The solutions to the $N-1$ ROPF were then checked for all line thermal limit constraints for both base-case and contingency-cases, identifying a set of the notable congested lines (NCL) that is then used as another source to identify critical line for solving future $N-1$ ROPF problems.

To assess the effectiveness of the proposed AHGNN's predictive capabilities, we also established four additional ML models, using GNN as the main architecture, for comparative analysis, examining the validity of their predictions and performance against the full $N-1$ OPF. These five ML models were created using a combination of various predictions for the base cases, contingency cases, and notable congested lines. The ideal model should have predictions that result in zero constraint violations for the $N-1$ ROPF solutions while the objective total costs should be nearly identical to the full

$N-1$ OPF solutions. With the proposed AHGNN model, a significant reduction in computational time required to find optimal solutions is achieved.

Furthermore, we delve into enhancements in hyperparameter tuning, especially loss attenuation during the training phase of the GNN model. We implemented a weighted factor (WF) to the loss function to limit predictions that are more likely to lead to violations of constraints. We perform sensitivity analysis to select the best weighted factor for the proposed GNN models. Predictions using the new loss function showed a remarkable decrease in the violation of constraints for the $N-1$ ROPF solutions.

3.2. Backgrounds

3.2.1. $N-1$ Optimal Power Flow

The OPF model is commonly used in practical power systems for real-time economic dispatch. It optimizes the generation dispatch points while satisfying multiple constraints throughout the transmission network and focuses on minimizing the total operating cost. For the OPF model presented below, the objective function is to minimize the total system generation cost as shown in

$$\mathbf{Obj} : \min \sum_{g \in G} c_g P_g. \quad g \in G \quad (3-1)$$

The generation limits as well as generation reserve requirements are described as follows:

$$\sum_n P_g^{res} = 0.05 \times \sum_n d_n \quad g \in G, n \in N \quad (3-2)$$

$$P_g^{min} \leq P_g \leq P_g^{max} - P_g^{res} \quad g \in G \quad (3-3)$$

$$P_g^{res} \geq 0 \quad g \in G \quad (3-4)$$

and that is set at 5% of the total load. Line flows are calculated in

$$P_k = \frac{\theta_{f(k)} - \theta_{t(k)}}{x_k} \quad g \in G \quad (3-5)$$

and are subject to line thermal capacity limits in

$$-RateA_k \leq P_k \leq RateA_k. \quad k \in K \quad (3-6)$$

The nodal power balance is enforced in

$$\sum_{g \in G(n)} P_g + \sum_{k \in K(n-)} P_k - \sum_{k \in K(n+)} P_k = d_n. \quad n \in N \quad (3-7)$$

To comply with the $N-1$ reliability requirement, the impacts of various single-element contingencies are evaluated and considered in OPF, creating the $N-1$ OPF model. To be consistent with the industry practices [41], [42], radial line contingencies that will lead to network islanding are not considered in the $N-1$ OPF model in this work. Constraints given below:

$$P_g^{min} \leq P_{cg} \leq P_g^{max} \quad c \in C, g \in G \quad (3-8)$$

$$-P_g^{ramp} \leq P_{cg} - P_g \leq P_g^{ramp} \quad c \in C, g \in G \quad (3-9)$$

are used to ensure the updated generators' power outputs under contingency are still within the associated limits and do not violate the ramping rate limits. Post-contingency line flow equations and thermal limit constraints for each $N-1$ contingency case are shown in:

$$P_{ck} = \frac{\theta_{f(ck)} - \theta_{t(ck)}}{x_k} \times N1_{ck} \quad c \in C, k \in K \quad (3-10)$$

$$-RateC_k \leq P_{ck} \leq RateC_k. \quad c \in C, k \in K \quad (3-11)$$

While the post-contingency nodal power balance for each $N-1$ contingency case is

$$\sum_{g \in G(n)} P_{cg} + \sum_{k \in K(n-)} P_{ck} - \sum_{k \in K(n+)} P_{ck} = d_n. \quad c \in C, k \in K \quad (3-12)$$

For practical independent system operators (ISOs), the line thermal limit constraints will be applied only on a small subset of lines that are heavily loaded or congested. These subsets of lines are chosen based on historical load profiles and line loading levels. By keeping an eye on these congested lines and zeroing in on the subsets of heavily loaded or congested lines, ISOs can ascertain that the power system network is operating efficiently while ensuring grid reliability. For $N-1$ OPF problems, the optimal solutions may be constrained by congested lines in both the base case and the $N-1$ cases. These congested lines are considered critical since they restrict the solution within a bounded feasible region. Uncongested lines are not monitored since the optimal solution is not affected by these variables.

By replacing (3-6) and (3-11) with:

$$-RateA_r \leq P_r \leq RateA_r \quad c \in C, r \in R \quad (3-13)$$

$$-RateC_r \leq P_{cr} \leq RateC_r \quad c \in C, r \in R \quad (3-14)$$

respectively, the updated $N-1$ OPF model can monitor only a small number of critical congested lines and ignore other non-critical uncongested lines in both the base case and contingency cases. As a result, the majority of line rating constraints could be eliminated from the OPF model, which will substantially reduce the model size and complexity.

3.2.2. Graph Neural Network: XENET layer

Machine learning is a well-established computer algorithm that has been extensively researched and applied in power system optimization. ML models have the remarkable ability to automatically improve and learn from vast amounts of data. These models are trained using data samples to make predictions or decisions without explicit programming. Among popular ML models are CNN and DNN, which have found broad applications across various disciplines. However, these models often overlook the importance of considering the network topology in their predictions.

For electrical networks, the topology contains crucial but implicit information and features that could significantly enhance the learning process. This is where the GNN model holds a distinct advantage over both CNN and DNN models. The GNN model utilizes an adjacency matrix (a matrix of size nb by nb , where nb represents the number of buses) to represent the network topology. Furthermore, the GNN model leverages the global context of the network to establish relationships between multiple node-level and edge-level features. In essence, the GNN model allows an optimizable transformation on all attributes of the graph, preserving graph symmetries, and permutation invariance.

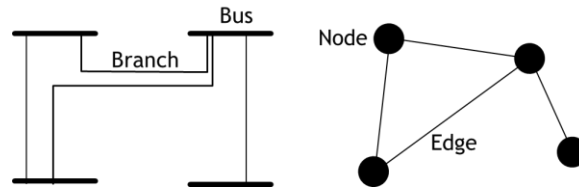


Fig. 3-1. Transformation of bus-branch model into a graph format.

Fig. 3-1 above illustrates the transformation of a simple power system's bus-

branch model into a graph format. Each bus in the network is represented as a node, holding information such as generation and load profiles as node features. Line flow and line thermal ratings are represented as edge features. The adjacency matrix effectively characterizes the connections between each bus and its neighboring buses. GNN has been used in diverse applications, serving various purposes in node-level, edge-level, and graph-level tasks, depicted in Fig. 3-2 below [19], [44], [45].

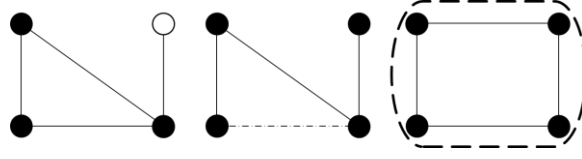


Fig. 3-2. Node-level vs. Edge-level vs. Graph-level classification task.

During the training stage of the GNN model, the equation below

$$h_i^k = \sigma(W_{self}^k \cdot h_i^{k-1} + W_{neigh}^k \cdot AGG(h_j^{k-1}, \forall j \in \Omega_i)) \quad (3-15)$$

describes the relationship between each node and its neighboring nodes in each forward pass of the training stage. h_i^k contains all the feature information of the current node, while W_{self}^k and W_{neigh}^k are node-wise shared weight matrices. $AGG(h_j^{k-1})$ is an aggregation function that combines feature information over all neighbor nodes, Ω_i . For each forward pass, the information acquired from each neighboring node is aggregated and passed through to the next training step. With a well-built GNN model with enough layers, feature information across the entire network can be utilized to enrich the prediction capability of the model. By leveraging the topology of the network during the training process, we can uncover valuable insights and improve the efficiency

of power system optimization through GNN.

For this work, the XENet layer was utilized as a graph convolutional layer and an integral part of the proposed GNN model. XENet layer was chosen due to its ability to convolve over both edge and node. The XENet layer is a message-passing layer that simultaneously accounts for both the incoming and outgoing information of neighboring nodes, such that a node's representation is based on the messages it receives as well as those it sends.

For a given graph, vector x_i contains the features associated with node i , and vector $e_{(i,j)}$ contains the features belonging to edge between node i and node j . The XENet layer aggregates information of the feature stacks s_{ij} in the following:

$$s_{ij} = \varphi^{(s)}(x_i \parallel x_j \parallel e_{(i,j)} \parallel e_{(j,i)}) \quad (3-16)$$

$$s_i^{out} = \sum_{j \in N(i)} a^{out}(s_{ij}) \cdot s_{ij} \quad (3-17)$$

$$s_i^{in} = \sum_{j \in N(i)} a^{in}(s_{ij}) \cdot s_{ji} \quad (3-18)$$

for message-passing between each forward pass. By concatenating the node and edge attributes associated with the incoming and outgoing messages of each node in (3-16), the multi-layer perceptron $\varphi^{(s)}$ learns to process information such that a node's representation is based on the messages it receives as well as those it sends. The feature stacks are used to aggregate incoming/outgoing information, using self-attention to compute a weighted sum, in (3-17) and (3-18) respectively. The incoming/outcoming messages are concatenated and used to update the node attributes, x'_i of the graph in

$$x'_i = \varphi^{(n)}(x_i \parallel s_i^{out} \parallel s_i^{in}). \quad (3-19)$$

Finally, some additional processing of the feature stacks through $\varphi^{(e)}$ in

$$e'_{(i,j)} = \varphi^{(e)}(s_{ij}) \quad (3-20)$$

allows us to compute new edge attributes, $e'_{(i,j)}$, that are dependent on the message exchange between nodes.

The architecture of the XENet layer is described using the given equations above where $\varphi^{(s)}$, $\varphi^{(n)}$, $\varphi^{(e)}$ are multi-layer perceptrons with parametric rectified linear unit as the nonlinear activations, and a^{out} and a^{in} are two dense layers with sigmoid activations and a single scalar output.

XENet is a critical part of our GNN model's architecture. XENet allows us to pay special attention to both incoming and outgoing edge attributes. It enables us to perform edge-level predictions across the network by stacking features of nodes and edges, and then passing those learned features through each forward pass. The proposed AHGNN method relies on the XENet layer to predict congested lines accurately in order to remove unnecessary constraints. It enables us to create a computationally efficient $N-1$ ROPF model that decreases the solving time to find the optimal solution.

3.3. Methodologies

This section explains the proposed AHGNN method that can dynamically determine the monitor lines for the base case and contingency cases: the offline training stage and online prediction stage are presented in two subsections separately. A detailed explanation of the data generation process is also provided. In addition, four benchmark

methods are also defined to gauge the proposed AHGNN-based critical line selection approach.

3.3.1. Data Generation

A large number of samples were generated to represent the historical grid data, which were used to train the proposed ML models. We ran $N-1$ OPF simulations on the test power system with different load profiles to collect enough samples, which is 10,000 in this work. The load profile of each sample is varied using two random variable parameters. The first parameter shifts the entire load profile up or down within a range of $\pm 5\%$ of the initial load profile. The second parameter shifts load profile at each individual bus within a range of $\pm 2\%$. The combination of both parameters provides varying load profile curves.

Once the samples had been generated, we solved full $N-1$ OPF and obtained the solution for each sample. From the solutions, we labeled line flows from all branches based on a pre-defined threshold: the lines whose line flows are above this threshold are considered critical, which will be included in the line monitor set for $N-1$ ROPF.

In the IEEE 73-bus system, there are a total of 108 branches, of which 2 branches are radial lines. Radial lines are branches that, once removed, will cause network separation. As illustrated in Fig. 3-3, the two red lines connecting nodes 30 and 54 (blue dot) are radial lines. In the full $N-1$ OPF model, all non-radial line contingency cases, in addition to the base case, are considered. Thus, the base case and 106 contingency cases are simulated and modeled in $N-1$ OPF problem. Using ML, congested lines will be predicted for each case of the $N-1$ ROPF problem.

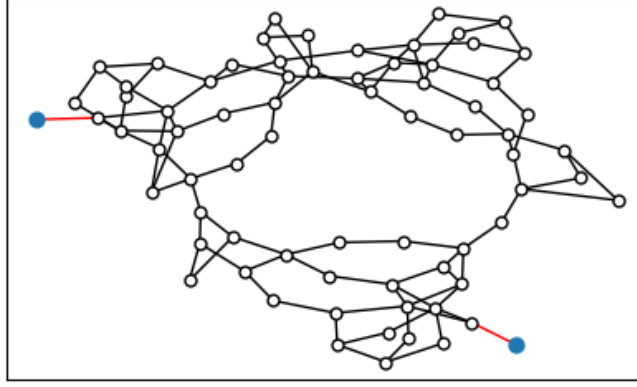


Fig. 3-3. Graph layout of IEEE 73-bus system.

From the initial 10,000 collected data samples, for the OFFLINE – Training mode, we separated it into three data set:

- Training data set for BC-GNN model
- Validation data set for BC-GNN model
- Testing data set for BC-GNN model.

The testing data set for BC-GNN is further split into two for:

- Training data set for CC-GNN model
- Validation data set for CC-GNN model

The following sections will go into further detail on how each data set is used to train separate GNN models for predictions of congested lines in base cases and contingency cases.

3.3.2. OFFLINE – Training Mode

Fig. 3-4 illustrates the flowchart for training the two proposed ML models: (i) BC-GNN model for classification of congested lines in the base cases and (ii) CC-GNN

model for classification of congested lines in the contingency cases. Once all the data samples are generated, processed, and labeled, they will be divided into three separate data sets for training, validation, and testing. During the OFFLINE training stage, these three data sets will be used to train and evaluate the BC-GNN model to identify critical lines to be monitored for the base case of OPF, where no contingency is considered as part of the input features.

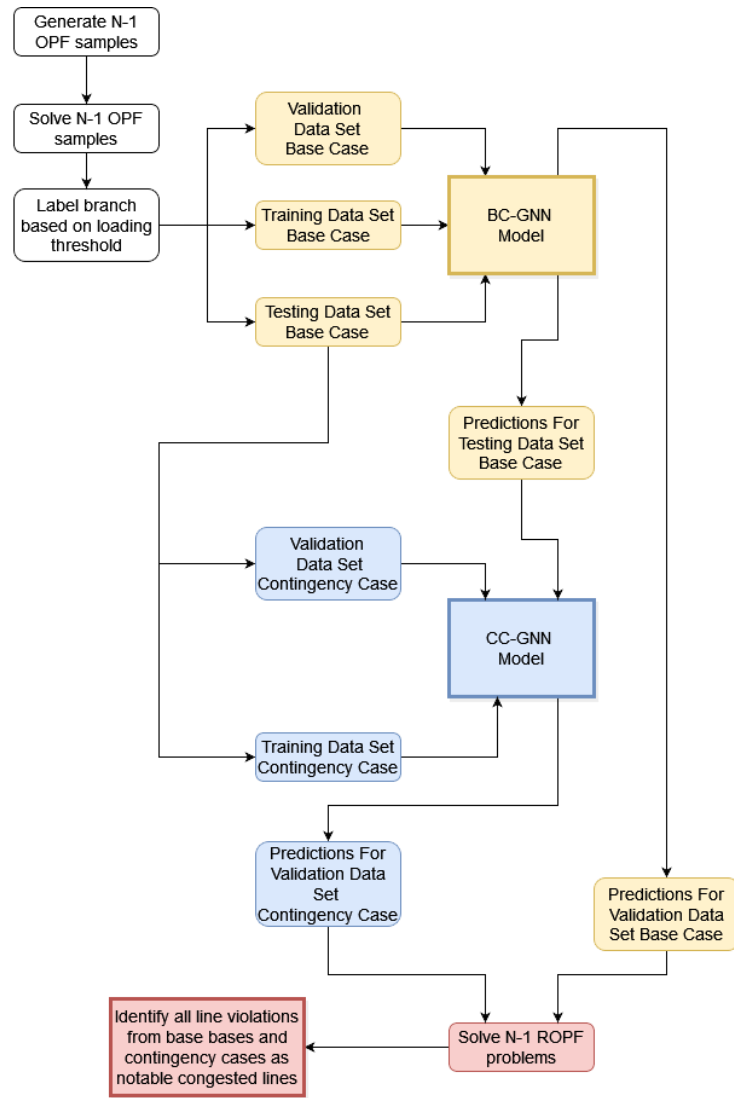


Fig. 3-4. Flowchart for the OFFLINE – Training mode.

Once the BC-GNN model is trained, prediction results from the testing data set are used as features for the CC-GNN model of the $N-1$ contingency cases. Since each sample contains data for both the base case and the contingency cases, samples from the BC-testing data set of the base cases are split into training and validation data sets of the contingency cases to train the CC-GNN model. By using output from BC-GNN as features for CC-GNN, we were able to enhance the prediction result of the second model CC-GNN. In essence, it is a nested two-stage training process. Finally, predictions are made on the $N-1$ cases validation data set using the trained CC-GNN model. These predictions combined with predictions made on the base cases validation data set are then used to solve $N-1$ ROPF problems. Congested lines from the solutions of $N-1$ ROPF problems are then identified as notable congested lines for the ONLINE prediction mode.

3.3.3. ONLINE – Prediction Mode

Once the models are trained, they can be deployed for online prediction. A separate data set of 1,000 samples that were never used during the training stage is used here to evaluate the effectiveness of both BC-GNN and CC-GNN models. The outputs generated by the BC-GNN model are incorporated as part of the features for the CC-GNN model. Furthermore, NCLs identified from past $N-1$ ROPF solutions during the OFFLINE training stage will also be included in $N-1$ ROPF (online) as monitor lines for new test samples.

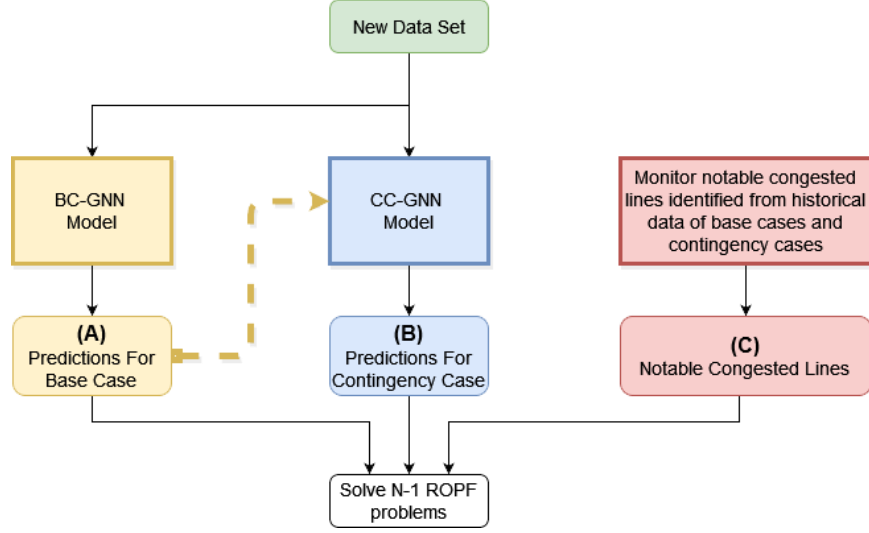


Fig. 3-5. Flowchart for the ONLINE - Prediction mode.

Fig. 3-5 summarizes how three distinct processes are used to identify critical lines, two use ML models and one uses previous solutions of $N-1$ ROPF, to only include possibly congested lines in the $N-1$ OPF problem. This strategic culling of uncongested lines plays a pivotal role in substantially reducing the number of constraints, consequently culminating in the enhancement of the $N-1$ ROPF model's efficacy. To gauge the effectiveness of our proposed approach, we establish a comparative benchmark by developing four other models for critical monitor line selection:

- Notable Congested Lines [NCL]: only use the notable congested lines identified offline [(C) as shown in Fig. 3-5].
- Graph Neural Network [GNN]: uses predictions from base case and $N-1$ cases, but without BC-GNN predictions as additional features for CC-GNN [(A) and (B), as shown in Fig. 3-5].

- Augmented Graph Neural Network [AGNN]: uses predictions from base case and $N-1$ cases [(A) and (B)], but without BC-GNN predictions as additional features for CC-GNN, along with notable congested lines [(C)].
- Hierarchical Graph Neural Network [HGNN]: uses predictions from base case and $N-1$ cases only, and with BC-GNN predictions as additional features for CC-GNN [(A) and (B)].
- **Augmented Hierarchical Graph Neural Network [AHGNN]** – the proposed method: uses predictions from base case and $N-1$ cases, and with BC-GNN predictions as additional features for CC-GNN, along with notable congested lines [(A), (B), (C)].

3.3.4. Loss Factor Function

Using an advanced GNN model for the prediction of line congestions, the uncorrected predictions will result in two possible types of errors: (i) type I false positive error – incorrectly predicting a lightly loaded line to be heavily loaded; and (ii) type II false negative error – incorrectly predicting a heavily loaded line to be lightly loaded. Having more type II errors will decrease the size of the subset of critical lines that we need to watch for $N-1$ ROPF, which in turn will likely speed up the solving time. However, incorrect predictions leading to type II errors have a direct correlation to the number of constraint violations in the final optimal solution. Thus, we want to enhance the GNN model to lower the number of type II errors, which in turn will dramatically reduce the number of constraint violations.

To improve the accuracy level of prediction, we need to significantly decrease the loss level during the training process.

$$Loss = -\sum_{i=1}^N y_i \times \log(\hat{y}_i) \quad (3-21)$$

describes the typical categorically cross-entropy loss function, where y_i is the truth label and $\hat{y}_i$ is the probabilities of the prediction. We introduce the term weighted factor, WF , in

$$WF = [1, \dots, R]. \quad R > 0 \quad (3-22)$$

It is a relative ratio between 0 and 1 for each class in the loss function. By multiplying the term WF in

$$WF \text{ Loss} = -\sum_{i=1}^N y_i \times \log(\hat{y}_i \times WF), \quad (3-23)$$

we created a new loss equation that emphasizes the loss value of one class over another. Essentially, the loss value will be propagated throughout the model and minimize the prediction error of type II over type I. By having fewer type II errors and more type I errors, the number of constraints in the $N-1$ ROPF model will increase, but the probability of having line constraint violations that lead to infeasible OPF solutions will significantly decrease.

3.4. Results and Analysis

GNN models and $N-1$ ROPF algorithm were developed in Python. We used the Pyomo library [20], [21] and Gurobi solver to solve $N-1$ ROPF problems. The GNN models were built using the spektral library [22]. All figures and graphs were generated

using matplotlib [46] and networkx [5]. A computer with 4 cores i7-4790K CPU, GTX3090 graphic card, and 32GB memory was utilized for all coding activities.

3.4.1. Model Training

From Fig. 3-6 and Fig. 3-7, the BC-GNN model performed well during the training phase. The curve for training loss tracked well with validation loss. The loss plot has a smooth decline that leads to a flat plateau. The percentage accuracy of prediction has a similar, but inverse appearance with a nearly 99% accuracy for the base case model. Both figures suggested that the training process went well where maximum accuracy is achieved while minimizing loss. The curves for loss plot and accuracy plot of CC-GNN model exhibit the same appearance as BC-GNN and are not shown here.

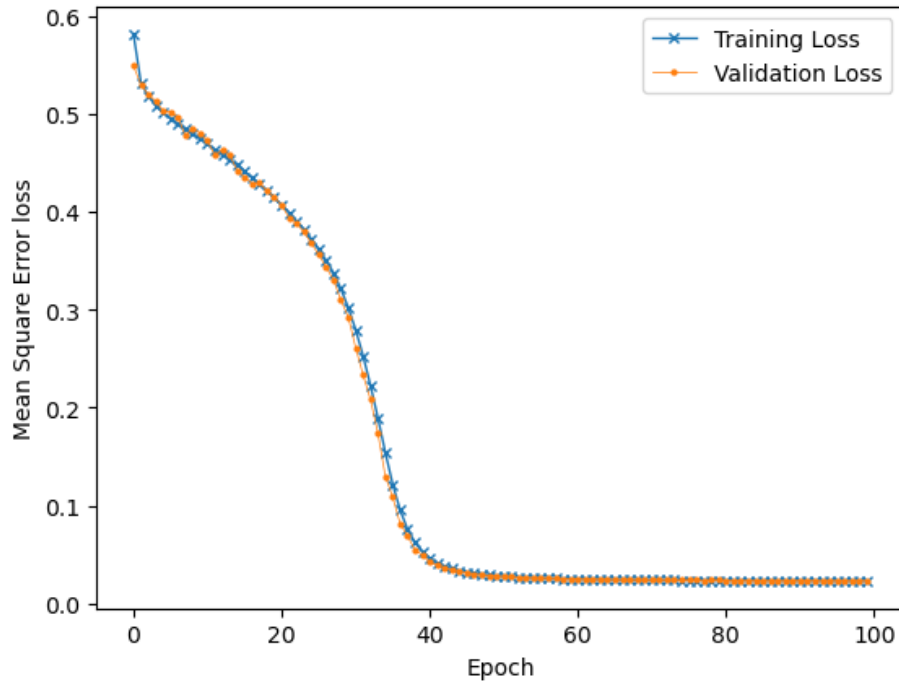


Fig. 3-6. MSE loss for training data set vs. validation data set during the training phase of the BC-GNN model.

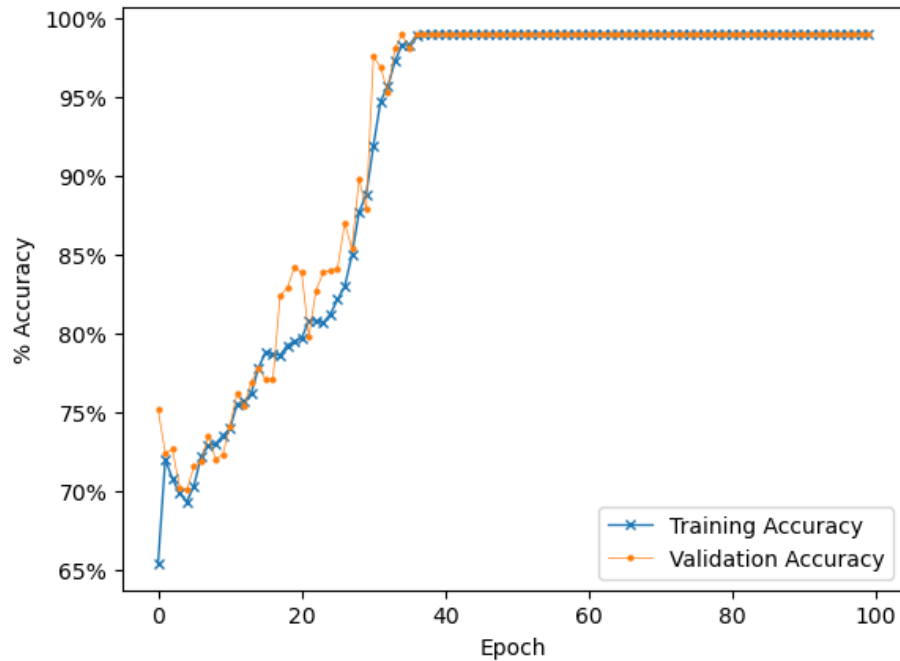


Fig. 3-7. Percent accuracy of prediction for training data set vs. validation data set during the training phase of the BC-GNN model.

3.4.2. Loss Factor function

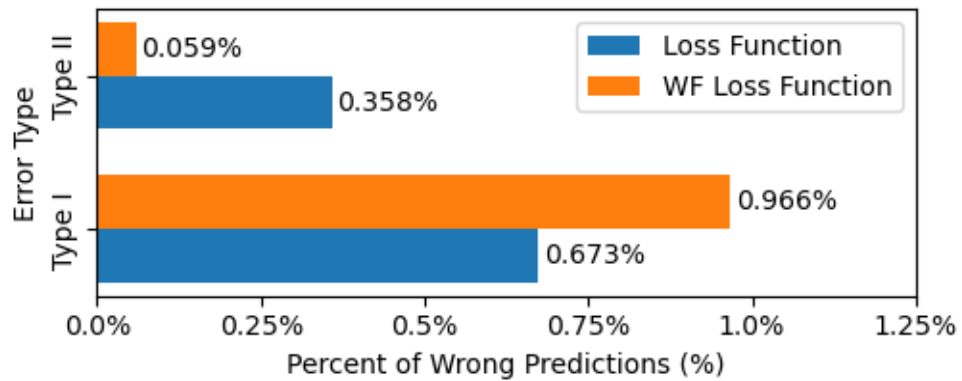


Fig. 3-8. Comparison of percentage error between type I and type II using the typical loss function vs. the (Weighed Factor) WF loss function.

By implementing the WF loss function, a significant decrease in type II errors within our baseline prediction was observed. As depicted in Fig. 3-8, the incidence of

type II errors dropped from 0.358% to 0.059%, constituting a remarkable six-fold reduction in magnitude. Nevertheless, the overall percentage of incorrect predictions, encompassing both type I and type II errors, remained relatively unchanged.

Fig. 3-9 illustrates our tracking of both error types across the first 50 epochs of the training process. Notably, while the type II error rate exhibited a high initial value, it substantially decreased in the initial 30 epochs and subsequently stabilized by the 50th epoch. This observation underscores the efficacy of our new loss function in adeptly propagating gradient loss during training, effectively curtailing the error rate pertaining to one specific class. Our hypothesis posits that the emphasis on the loss value of one class over the other facilitated a reduction in prediction errors for that class while keeping the total error rate at the same level. The outcomes of our analysis affirm our success in markedly mitigating type II error predictions, consequently leading to a lower frequency of line rating violations within the $N-1$ ROPF solutions.

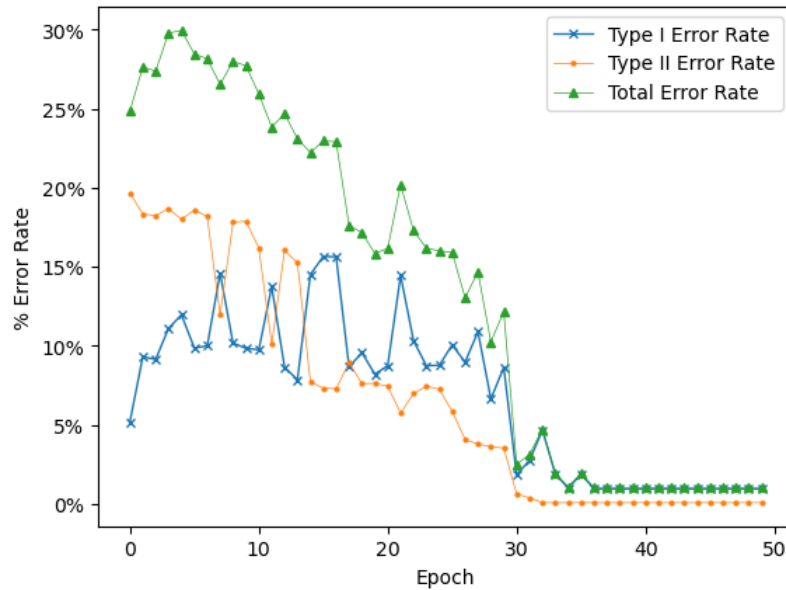


Fig. 3-9. Percent of error rate during the training phase of the BC-GNN model.

3.4.3. The Proposed AHGNN Model

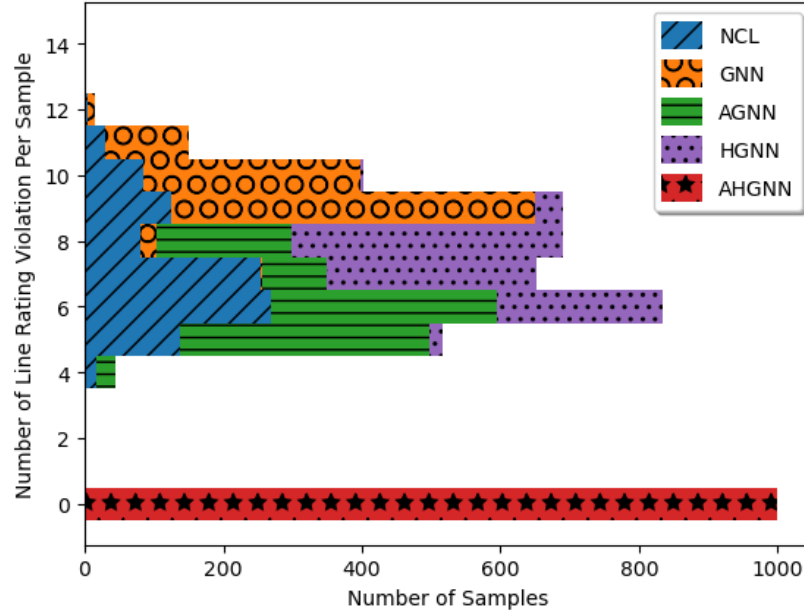


Fig. 3-10. Comparison of the number of line rating violations for the five models.

Upon obtaining predictions for line congestion, these forecasts are integrated into the $N-1$ ROPF problem. Once solutions have been found for the $N-1$ ROPF problem, a thorough validation process was used to ensure that all stipulated constraints have been met. Remarkably, our proposed model, AHGNN, outperformed all the other benchmark models by a significant margin. This superior performance is illustrated in Fig. 3-10, where we present a stacked bar chart that contrasts the incidence of line rating violations per sample within the $N-1$ post-contingency scenarios, utilizing a data set of 1000 samples. Among the five models examined, it is noteworthy that only the AHGNN model boasts zero line rating violation across all solution instances for each sample.

TABLE 3-1 - Line rating violations for the five proposed models per 1000 samples.

Number of Line Rating Violations Per Sample		NCL	GNN	AGNN	HGNN	AHGNN
	0	0	0	0	0	1000
	1	0	0	0	0	0
	2	0	0	0	0	0
	3	0	0	0	0	0
	4	16	0	28	1	0
	5	138	0	359	19	0
	6	269	0	326	239	0
	7	254	3	92	304	0
	8	81	23	195	391	0
	9	124	525	0	42	0
	10	84	314	0	4	0
	11	29	122	0	0	0
	12	5	10	0	0	0
	13	0	2	0	0	0
	14	0	1	0	0	0
Total		1000	1000	1000	1000	1000

The four alternative benchmark models encountered challenges in effectively solving $N-1$ ROPF solutions without violations. A detailed breakdown of the frequency of line rating violations for each model is presented in TABLE 3-1. Among the other four benchmark models, instances of $N-1$ ROPF solutions consistently violated line

rating constraints were observed, ranging from a minimum of 4 lines to a substantial number of 14 lines. The tabulated data strongly suggests that employing isolated prediction sources, without integration, fails to yield solutions of sufficient quality that adhere to all modeling constraints. Notably, within the GNN model, reliance solely on predictions from the base and $N-1$ cases is associated with the highest count of line rating violations compared to other models. This is particularly evident in over 500 samples that experience 9 or more line rating violations. Despite the incorporation of base case predictions to enhance $N-1$ predictions in the HGNN model, the solutions still fall short of desired expectations.

Conversely, the AHGNN model, which incorporates NCL into its predictive process, stands as the sole exception. This model consistently generates solutions free from any constraints or line rating violations, underscoring its effectiveness and superiority in producing robust and accurate $N-1$ ROPF solutions.

In Fig. 3-11, we employed a scatter plot to display the objective total cost of the five proposed $N-1$ ROPF models against the objective total cost of the full $N-1$ OPF model. To enhance clarity without overwhelming the plot, we have confined the visualization to the initial 100 samples from our data set.

Notably, in Fig. 3-11 , part (ii) and (iv), both the GNN model and HGNN model exhibit total costs lower than those of the full $N-1$ OPF model. This outcome comes from the breach of constraints in the $N-1$ scenarios. For the remaining models, the overall objective cost is around 0.25% higher in comparison to the full $N-1$ OPF model, a foreseeable result. This divergence can be attributed to our reduction of line constraints during the $N-1$ OPF formulation stage, which consequently relaxes the constraints

governing generation ramping rates in the $N-1$ post-contingency scenarios. It is important to note that these ramping rates have an impact on the total cost of each sample.

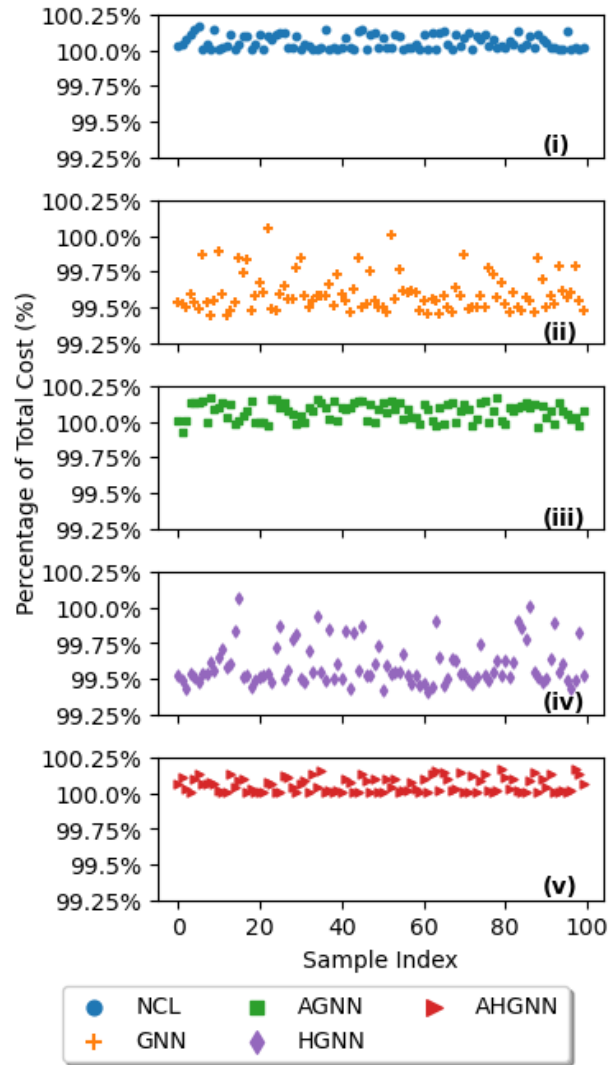


Fig. 3-11. Comparison of the objective total cost for the five proposed models.

Despite the variance in the total objective cost from the full $N-1$ OPF model, we truly believe that modest increment has an inconsequential influence on the integrity of the $N-1$ ROPF solutions. As TABLE 3-2 outlines, the proposed AHGNN model

manifests a maximum total cost of 100.193% compared to the full solution, while the median objective total cost stands at 100.043%. It signifies that the majority of $N-1$ ROPF solutions using AHGNN model contain an objective total cost with negligible increase compared to the full $N-1$ OPF solutions.

TABLE 3-2 - Statistical data for objective total cost for the five proposed models.

Model	NCL	GNN	AGNN	HGNN	AHGNN
Mean	100.062	99.594	100.056	99.594	100.061
Median	100.043	99.551	100.046	99.551	100.043
Max	100.193	100.105	100.202	100.077	100.193
Min	100.002	99.365	99.919	99.365	100.002
Std.	0.052	0.135	0.060	0.135	0.051

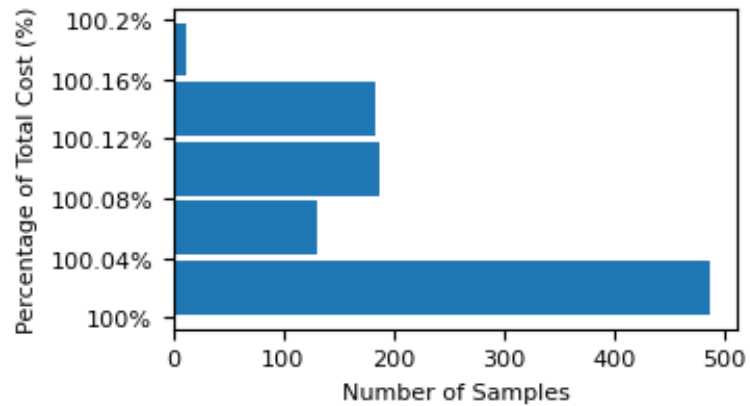


Fig. 3-12. Histogram of the total objective cost for AHGNN model.

Our assertion is validated in Fig. 3-12, where nearly 500 solutions register an objective total cost increase of less than 0.04%—a marginal and insignificant deviation.

3.4.4. Solving Time Performance

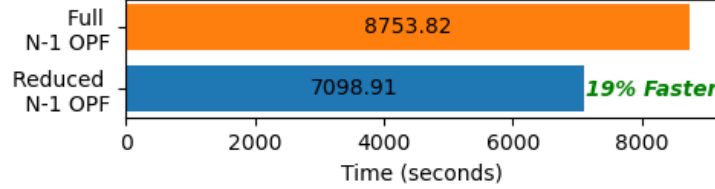


Fig. 3-13. Comparison of solving time between full $N-1$ OPF and $N-1$ ROPF.

In the preceding sections, we conducted a comprehensive comparison of five benchmark models to evaluate their performance and efficacy in adhering to all constraints while staying close to the total objective cost. In the current section, we shift the focus to the computational performance analysis of our proposed model, AHGNN. Given that only the $N-1$ ROPF solutions obtained using the AHGNN model exhibit no line rating violations, these are the sole results that can be directly compared to the solutions of the full $N-1$ OPF.

To facilitate a comprehensive assessment, we solved the same data set of 1000 samples using both the $N-1$ ROPF method and the full $N-1$ OPF method across five separate runs. Averaging the outcomes, we present the mean results in Fig. 3-13. The visualization showcases the total amount of time required to solve all 1000 samples. Specifically, the $N-1$ ROPF method demonstrated its capability by solving these 1000 samples in 7,098 seconds, reflecting a noteworthy 19% enhancement in speed as compared to the full $N-1$ OPF method.

While the reduction in the number of constraints was as high as 26%, going from around 84,000 down to 64,000, the actual decrease in computing time across all samples didn't exhibit a similar margin. The phenomenon can be attributed to the small size of the IEEE 73-bus system, containing just 108 branches. Consequently, the reduction in the number of constraints is not large enough, yielding a relatively proportional lesser decrease in computing time. However, it is worth noting that in larger systems with thousands of buses and a substantially higher number of branches, the gains in computing time reduction will likely rise exponentially.

This research paper specifically concentrates on the pragmatic application of the AHGNN model to expedite $N-1$ OPF computing times. The observed 19% reduction, though modest in percentage terms, is a significant achievement that aligns with our expectations and goals.

3.5. Conclusion

Our research has significantly reduced the number of constraints in the comprehensive $N-1$ ROPF problem. We achieved this by using the proposed innovative AHGNN model to formulate a streamlined $N-1$ ROPF algorithm that combines predictions from three distinct sources. This strategic integration of predictions resulted in a remarkable 19% decrease in solving time, while still preserving the quality of the solutions. We believe that by adopting a diverse strategic approach, we have designed an AHGNN model that can expedite solving times for $N-1$ ROPF problems.

For our current research, we focused on a relatively modest IEEE 73-bus system. However, we envisioned our research will be applied towards a larger-scale system, one

that encompasses thousands of buses. We hypothesized that such a shift could result in substantial computational resource savings, a prospect that fuels our anticipation for the future trajectory of our research. For future projects, we plan to explore alternative topologies for GNN, with the ultimate goal of enhancing our predictive capabilities. Our methodology involves the utilization of diverse topologies for each bus, allowing us to intricately model each load and generator separately, thus achieving a more nuanced and detailed representation of the system dynamics. Looking forward, we plan to seamlessly integrate GNN into other advanced OPF problems, for instance, network reconfiguration OPF, thereby broadening the scope and applicability of GNN in power systems. This strategic expansion promises to unlock new insights and solutions in the realm of power system optimization, reinforcing the significance of ML technologies especially GNN.

4. GRAPH NEURAL NETWORK-ACCELERATED NETWORK-RECONFIGURED OPTIMAL POWER FLOW

Optimal power flow (OPF) has been used for real-time grid operations. Prior efforts demonstrated that utilizing flexibility from dynamic topologies will improve grid efficiency. However, this will convert the linear OPF into a mixed-integer linear programming network-reconfigured OPF (NR-OPF) problem, substantially increasing the computing time. Thus, a machine learning (ML)-based approach, particularly utilizing graph neural network (GNN), is proposed to accelerate the solution process. The GNN model is trained offline to predict the best topology before entering the optimization stage. In addition, this paper proposes an offline pre-ML filter layer to reduce GNN model size and training time while improving its accuracy. A fast online post-ML selection layer is also proposed to analyze GNN predictions and then select a subset of predicted NR solutions with high confidence. Case studies have demonstrated superior performance of the proposed GNN-accelerated NR-OPF method augmented with the proposed pre-ML and post-ML layers.

4.1. Literature Review

Transmission networks are highly meshed networks since they are critical electricity infrastructure that was built with redundancy. This indicates there are usually multiple electrical paths for long-distance power transfer from one area to another, enabling flexible transmission with dynamic network topologies. However, transmission networks have been traditionally considered static in real-time grid operations in the power system industry. For instance, optimal power flow (OPF) that

is used to perform real-time generation redispatch does not model network reconfiguration. While traditional OPF only conducts generation-side regulation, the exploration of extra topology-aware regulation mechanisms in OPF will improve the grid efficiency and reliability.

Large power systems often face congestion, especially during peak times. Building new transmission lines is costly and time-consuming. Network-reconfigured optimal power flow (NR-OPF) is a solution that optimizes the existing network layout and power flows throughout the grid to make it more efficient. NR-OPF, also referred to as optimal transmission switching (OTS), can reduce network congestion, allowing for more renewable energy and lowering the total cost. It offers benefits like improved reliability, lower losses, and better voltage control [47]. Essentially, NR-OPF helps make the power grid more efficient and adaptable.

NR-OPF has been widely researched since its introduction. It can be applied to both short-term operations and long-term planning, at both the transmission and distribution level [48]. Importantly, it helps integrate large amounts of renewable energy into the grid. NR-OPF strategically changes the network layout by flexibly switching transmission lines on or off. This improves power flow distributions and makes better use of existing infrastructure. By identifying which lines to switch, grid operators can relieve lines congestion, improve voltage, and make the system more reliable [49].

NR-OPF can address suboptimal power system performance resulting from variability in load and generation patterns. Different scenarios of power generation and load profile may require distinct network topologies to achieve optimal efficiency. Consequently, a network optimized for one set of conditions may perform sub optimally

when those conditions change [50]. Extending linear programming (LP) OPF problems, NR-OPF problems are commonly modeled as mixed-integer linear programming (MILP) problems. These formulations utilize binary variables and the big-M method for linearization, where the binary variables represent the switching status of each transmission line.

For large-scale networks with numerous transmission lines, the combinatorial explosion of binary variables significantly increases the computational complexity, leading to extended solution times. It is computationally demanding to determine the switching status of each transmission line in NR-OPF problems [51], [52]. Despite the substantial advantages NR-OPF offers over traditional OPF methods, these MILP-based NR-OPF problems pose significant challenges. The large and complex solution space makes it difficult to guarantee a globally optimal solution. The combinatorial nature of network topology—where the number of possible configurations grows exponentially with network size—further exacerbates the complexity of the problem [53].

The high computational complexity is a major bottleneck preventing the industrial adoption of the enhanced NR-OPF as a real-time application. Traditional optimization methods often struggle to handle this rapid increase in complexity, leading to potentially long solving times and suboptimal solutions. This inherent difficulty underscores the need for innovative and efficient approaches to tackle these challenging optimization problems in the power systems domain. Therefore, enhancing the computational efficiency of the NR-OPF problem is paramount. Reducing the computational complexity of NR-OPF problems can be achieved by strategically limiting the number of switchable lines considered.

In [54], a method is proposed to identify and rank the constraints that actively shape the OPF feasible region, based on their influence on the cost function. A heuristic algorithm is then developed that iteratively removes these constraints and solves a series of standard OPF problems, leading to more efficient solutions. In [55], different criteria concerning congestions, limit violations and production costs are used to identify candidate lines for switching. Machine learning (ML), specifically reinforcement learning and graph neural network (GNN), has been used in [56] to develop a heuristic algorithm for OTS at the distribution level. A new method for managing overloaded transmission lines during contingencies is offered in [57] to change the network's structure using NR-OPF. It introduces a two-stage optimization model that uses a DC model for an initial solution followed by an AC model for further optimization. Previous approaches haven't fully leveraged the potential of diverse system topologies or adapted to varying load profiles when identifying switching lines for transmission systems. There is a need for an adaptable model that can swiftly modify system layouts, leading to enhanced efficiency and reliability in power grids.

In this paper, we focused on utilizing ML, especially GNN, as an alternative method to speed up the NR-OPF solving process and address the drawbacks of existing works. Graph neural networks have shown promise in various power system applications such as fault detection [29], time-series prediction [16], and power flow calculation [15]. Researchers are actively investigating different GNN types to further enhance power system control and optimization. GNN was explored to reduce the number of constraints in base-case OPF problems [35]. For $N-1$ OPF, GNN was used in [58] to predict both base-case and contingency-case line congestions and remove non-

congested lines in the monitoring set to reduce the model size and speed up the computing time. However, these methods are not applicable for NR-OPF, a much more complex MILP problem, due to the condition-based optimal dynamic topology scheme with NR leading to varying congestion bottlenecks. There were very limited studies on accelerating NR-OPF using ML. To bridge the gap, a novel GNN-accelerated NR-OPF (GaNR-OPF) approach with unique pre-ML filter and post-ML selection steps is developed in this paper to reduce NR-OPF model complexity, accelerate the solving process and thus reduce the computing time with quality solutions.

The contributions and innovations of this work are summarized as follows:

- First, we developed and trained a full GNN (FGNN) model to determine the optimal topology by predicting the switching status of each transmission line before entering the optimization stage of solving NR-OPF.
- Second, we developed an offline pre-ML filter step to identify non-critical lines whose switching status remains the same irrespective of the system loading levels, such that these non-critical lines will not be considered, leading to a reduced GNN (RGNN) model with less training time. The proposed RGNN model focuses the prediction process exclusively on the critical lines that influence the system's topology optimization decisions, streamline the utilization of ML technologies, and further improve its accuracy compared to the FGNN model.
- When adopting the predictions of FGNN or RGNN, it will convert the MILP NR-OPF into an LP OPF model with no more binary variables; however, it would result in increased cost or even infeasibility. To address this issue, we

developed a fast online post-ML selection step to analyze the GNN predictions and then only select a subset of predicted NR solutions with high confidence for adoption. The selection process ensures that the final selected predictions are as reliable as possible, contributing to the overall robustness of the GaNR-OPF model. With this post-ML step, the binary variable representing the switching status of selected lines will be fixed before entering the optimization stage. The full NR-OPF is transformed into a reduced NR-OPF. The reduced NR-OPF will still be a MILP problem, but it will have much fewer binary variables, ensuring solution quality while still reducing the computing time substantially.

- To compare against the traditional full NR-OPF model, we have formulated a total of four methods using GaNR-OPF model: (i) FGNN-LP with neither the pre-ML layer nor the post-ML layer, (ii) FGNN-MILP with the post-ML layer, (iii) RGNN with the pre-ML layer, and (iv) the proposed RGNN-MILP with both the pre-ML and post-ML layers. Case studies have demonstrated RGNN-MILP method outperforms other benchmark methods. GaNR-OPF model, using RGNN-MILP, produces solutions with identical objective total costs to the full NR-OPF solutions and without any constraint violations. The proposed RGNN-MILP method demonstrated a substantial decrease in the computational time needed to find optimal solutions.

The remaining sections of this chapter are organized as follows. Section 4.2.1 provides a comprehensive introduction to NR-OPF. Section 4.2.2 introduces GNN and explores the architectural layers. Section 4.3 explains the methodologies employed in

our research. Section 4.4 presents a detailed analysis of the simulation. In Section 4.5, we draw conclusions based on our findings, summarize the key takeaways and discuss the implications of our research.

4.2. Backgrounds

4.2.1. Network-Reconfigured Optimal Power Flow

Network reconfiguration is a powerful concept incorporated into the OPF problem, empowering system operators with the ability to strategically and temporarily disconnect specific transmission lines from the network. This integration allows for the simultaneous optimization of both the grid's topology and the generation dispatch, ensuring a balance between operational efficiency and maintaining the system's overall reliability.

For the given traditional OPF problem below, with a bus set N , a branch set K , and a generator set G , the objective function is to minimize the total system generation cost as shown in

$$\mathbf{Obj} : \min \sum_{g \in G} c_g P_g \quad g \in G \quad (4-1)$$

in which P_g denotes the dispatched generation from each generator unit g while c_g denotes its associated operation cost. The following constraints:

$$\sum_n P_g^{res} = 0.05 \times \sum_n d_n \quad g \in G, n \in N \quad (4-2)$$

$$P_g^{res} \geq 0 \quad g \in G \quad (4-3)$$

set the generation reserve level, P_g^{res} , as 5% of the total amount of loads, d_n .

Generation output must meet generation limits, P_g^{min} and P_g^{max} , as well as generation reserve in

$$P_g^{min} \leq P_g \leq P_g^{max} - P_g^{res}. \quad g \in G \quad (4-4)$$

Line flows, P_k , are calculated as

$$P_k = \frac{\theta_{f(k)} - \theta_{t(k)}}{x_k} \quad k \in K \quad (4-5)$$

based on the reactance of the line, x_k , and the angle of the bus ($\theta_{f(k)}$ and $\theta_{t(k)}$). Line flows are also subjected to line limit ratings, $RateA_k$, in

$$-RateA_k \leq P_k \leq RateA_k. \quad k \in K \quad (4-6)$$

The nodal power balance for each node is enforced with

$$\sum_{g \in G(n)} P_g + \sum_{k \in K(n-)} P_k - \sum_{k \in K(n+)} P_k = d_n. \quad n \in N \quad (4-7)$$

The application of the big-M method with the equations below

$$-M(1 - NR_k) \leq P_k - \frac{\theta_{f(k)} - \theta_{t(k)}}{x_k} \quad g \in G, k \in K \quad (4-8)$$

$$P_k - \frac{\theta_{f(k)} - \theta_{t(k)}}{x_k} \leq M(1 - NR_k) \quad k \in K \quad (4-9)$$

facilitates linearization by introducing large negative constants associated with the constraints. These constants, denoted by M , are carefully chosen to ensure that they would not be part of any optimal solution if such a solution exists. This approach penalizes any violation of the constraints, effectively discouraging the optimization process from considering solutions that breach these limits.

In the NR-OPF problem, lines are selectively disconnected to improve the

economic efficiency of the power system. Equation for line flows in (4-5) are replaced by (4-8) and (4-9). The variable NR_k is a binary variable that describes the status of each line, where a value of 1 means ON and a value of 0 means OFF. Line limit ratings in (4-6) are multiplied with the variable NR_k to produce the equation below

$$-NR_k \times RateA_k \leq P_k \leq RateA_k \times NR_k, \quad k \in K \quad (4-10)$$

Moreover, this technique is commonly employed in formulating linear optimization problems where constraint violations incur a substantial positive penalty constant, again represented by M . The magnitude of M is crucial; it must be sufficiently large to render any constraint-violating solution significantly less attractive than feasible solutions [59], [60]. The big-M method plays a vital role in addressing non-linear constraints and ensuring constraint satisfaction within the optimization of the NR-OPF problem, thereby enhancing the effectiveness and reliability of the solution process.

In the context of NR-OPF, the NR vector transforms a linear programming problem into a more complex mixed-integer linear programming (MILP) problem. By strategically reducing the size of this NR vector, either through preprocessing techniques or by leveraging machine learning predictions, the overall complexity of the problem is significantly diminished. This simplification directly translates to faster computation times when seeking a solution.

4.2.2. Graph Neural Network: Optimal Topology

Machine learning algorithms, particularly convolutional neural networks (CNN) and deep neural networks (DNN), have been extensively utilized for power system optimization [9]. However, these models often neglect the inherent topological

information within electrical networks, which could significantly enhance their learning process. GNN addresses this limitation by incorporating network topology through an adjacency matrix, enabling a more comprehensive understanding of node and edge features. This research gap will be bridged in this paper by developing a GNN-based acceleration approach to reduce the NR-OPF complexity and thus the computing time.

The unique capabilities of GNNs suggest they have great potential to enhance existing power system optimization tools, such as finding faster solutions for OPF. By effectively capturing the complex interdependencies within a power system network, GNNs could lead to more accurate predictions of important variables like load and generation. Ultimately, this could translate to faster and more efficient solutions for complex optimization problems in the power industry.

In electrical networks, buses are represented as nodes with associated generation and load profiles, while line flows and thermal limits are represented as edge features. The adjacency matrix defines the connectivity between buses. GNN leverages this global context to establish relationships among multiple node and edge features, facilitating optimizable transformations while preserving graph symmetries and permutation invariance.

GNNs have diverse applications in node-level, edge-level, and graph-level tasks. Graph-level tasks involve determining a single property for the entire graph, such as predicting properties of protein molecules based on their molecular structure. Edge-level tasks identify associations between nodes, as exemplified by analyzing citation networks to understand connections and quantify the value of edges (citations). By incorporating network topology, GNNs offer enhanced capabilities for power system

optimization and various other domains where graph-structured data plays a crucial role.

GNN is specifically designed to work with network-structured data, as they can incorporate the network's topology into its input feature vectors. This allows them to understand and capture the dependencies and relationships that exist within complex graphs through a process called message passing between nodes. These message passing interactions allow information to flow across different layers of the GNN, making them especially effective for power system applications.

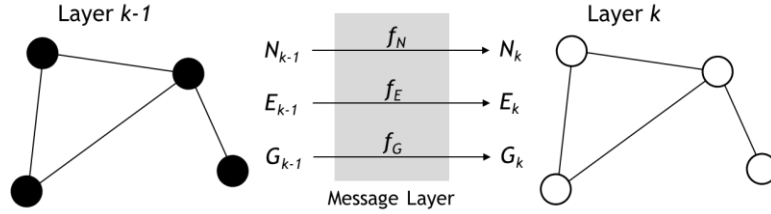


Fig. 4-1. Representation of GNN where node features N , edge features E , and graph features G are updated through a message layer.

In this study, the XENet layer was employed as a graph convolutional layer within the proposed GNN model. Its selection was motivated by its capability to perform convolution operations on both edge and node tensors, as demonstrated in [23]. The XENet layer functions as a message-passing layer, concurrently considering both incoming and outgoing information from neighboring nodes. This ensures that a node's representation is influenced not only by the messages it receives but also by the messages it sends. The XENet architecture layer has been mathematically described in section 3.2.2 of the previous chapter.

XENet plays a crucial role in our GNN model's architecture, allowing us to give special consideration to both incoming and outgoing edge attributes. By stacking

features of nodes and edges and passing these learned features through each forward pass, it enables us to perform edge-level predictions across the network. Our proposed GaNR-OPF model relies on the XENet layer to accurately predict the switching status of each transmission line. It allows us to simplify the problem and create a computationally efficient NR-OPF model, thereby reducing the time required to find the optimal solution.

4.3. Methodologies

This section details the GaNR-OPF model, a novel approach for dynamic line switching status determination. The model is divided into two distinct stages: offline training stage, and online prediction stage.

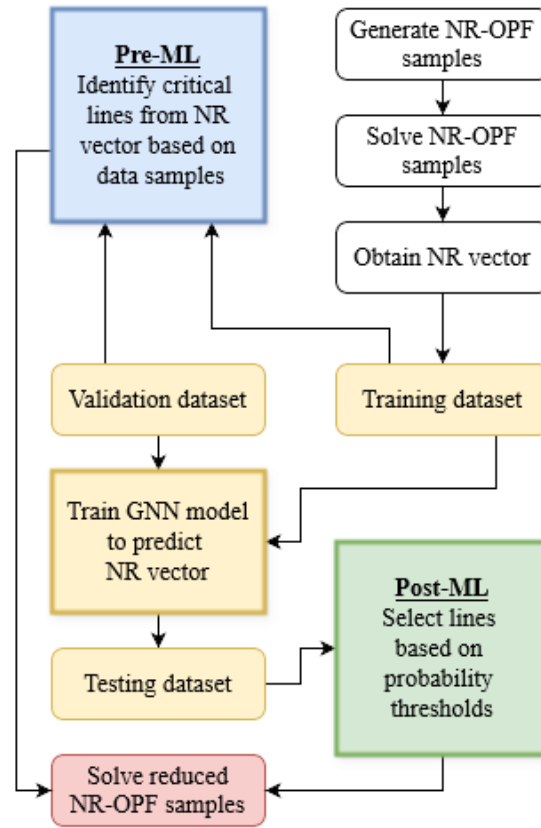


Fig. 4-2. Flowchart for the OFFLINE – Training Stage.

4.3.1. OFFLINE – Training Stage

This stage involves the development and training of the underlying GNN model that will be used for prediction. The specifics of the training process, including data preparation, model selection, and optimization.

Initially, a substantial dataset was generated to simulate historical grid data, serving as the foundation for training the proposed GNN models. This involved running 10,000 full NR-OPF simulations on the IEEE 73-bus system, each with load profiles varying within $\pm 7\%$ of the initial profile. These simulations yielded the NR vector, indicating the ON/OFF status of each transmission line, which was then used as the training output for the GNN model.

As shown in Fig. 4-2, the data samples were partitioned into three distinct datasets: training, validation, and testing. The training dataset was specifically used to train the GNN model to predict line switching statuses. Both the training and validation datasets were instrumental in preparing the pre-ML filtering step. Predictions derived from the testing dataset were then evaluated in the post-ML selection step. Finally, a selected subset of lines was utilized to solve the NR-OPF problem, and the resulting solutions were rigorously checked for validity and constraint satisfaction.

Pre-ML Filtering Step: The validation dataset was leveraged to analyze the NR vector and pinpoint "critical lines." These are defined as lines that consistently exhibit either a 100% ON or 100% OFF switching status across all the samples. Fig. 4-3 provides a visual representation of the line switching statuses derived from 10,000 NR-OPF solutions. A colormesh format is used where light (yellow) pixels signify an ON status and dark (purple) pixels indicate an OFF status.

By examining this graph, lines that remained either connected or disconnected throughout all the solutions were identified. These lines can be deemed non-critical as their status remains unaffected by varying loading conditions within the system. Consequently, these non-critical lines were removed from the GNN model's prediction process, resulting in a reduced GNN model. Strategic exclusion of these non-critical lines allows the GNN model to concentrate solely on predicting the switching status of the critical lines, those that are influenced by changes in loading conditions. This targeted approach not only streamlines the model's prediction task but also considerably improves its accuracy and efficiency by focusing on the most relevant lines within the IEEE 73-bus system.

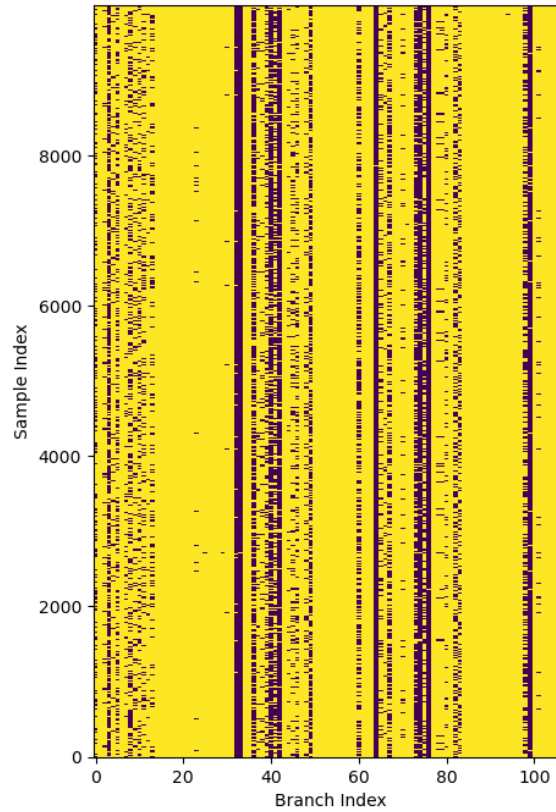


Fig. 4-3. Line switching status for 10,000 NR-OPF solutions data sample.

Post-ML Selection Step: The GNN model generated predictions for each line's switching status, expressed as probabilities of being ON or OFF. These probabilities summed to 100% for each line. To refine these predictions, a filter was applied: the lines with predicted probabilities between a lower limit of 5% and an upper limit of 95% were **Excluded**. These lines, in gray shading in TABLE 4-1, represent those where the GNN model is less confident about their switching status. Lines with probabilities outside this range, indicating high confidence in either the ON or OFF state, were **Selected** to have fixed status before entering the optimization stage, which enables to solve a reduced NR-OPF problem rather than the full NR-OPF problem. The switching status of the remaining **Excluded** lines is determined by solving the reduced NR-OPF. This approach focuses the optimization process on the lines where the GNN model's predictions are less certain, potentially leading to a more efficient and accurate solution. With post-ML step, the NR-OPF problem would become a reduced NR-OPF problem, still MILP but much smaller in terms of complexities.

TABLE 4-1 - Examples of probabilities of prediction for a few lines in percentage.

Line	1	2	3	4	5	6
OFF	6.11%	0.04%	7.53%	81.54%	23.03%	0.15%
ON	93.89%	99.96%	92.47%	18.46%	76.97%	99.85%
Post-ML Selection	Excluded	Selected	Excluded	Excluded	Excluded	Selected

4.3.2. ONLINE – Prediction Stage

In this operational stage, the trained GNN model is deployed to analyze real-time data and predict the optimal switching status for each line. To assess the effectiveness of the GaNR-OPF model, four benchmark methods are defined and implemented. Comparative analysis of these benchmarks will be conducted to evaluate the performance and advantages of the proposed method in terms of accuracy and efficiency.

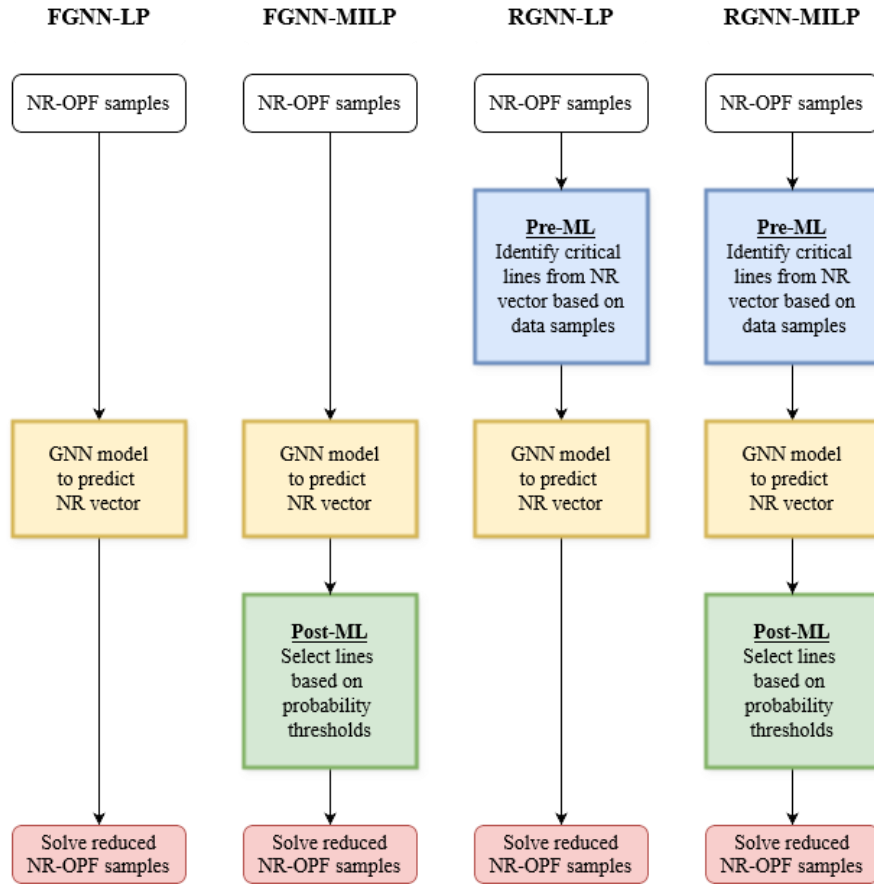


Fig. 4-4. Flowchart for the ONLINE – Prediction Stage using the four proposed benchmark methods.

Below are the four deployed methods:

- Benchmark FGNN-LP method: uses only a full GNN model for prediction of switching status of all lines without any extra processing steps.
- Benchmark FGNN-MILP method: uses the post-ML selection step, after using a full GNN model for prediction of switching status of all lines, to convert selected line switching binary variables into parameters.
- Benchmark RGNN-LP method: uses the pre-ML filtering step to train a reduced GNN model for prediction of switching status of a subset of critical lines.
- **Proposed RGNN-MILP method:** uses both the pre-ML filtering step and post-ML selection step.

Fig. 4-4 summarizes the flowchart for the four proposed methods. Once a smaller, more refined set of candidates for switching lines has been identified, these lines are then incorporated into the subsequent stage of the process: finding a solution for the reduced NR-OPF problem. By focusing on this smaller subset, the computational complexity of the NR-OPF problem is significantly reduced, potentially leading to faster and more efficient solutions.

4.4. Results and Analysis

GNN models and NR-OPF algorithms were written in Python. We used the Pyomo library [20], [21] and gurobi solver to solve NR-OPF problems. The GNN models were built using the spektral library [22]. All figures and graphs were generated

using matplotlib [46] and networkx [61] library. A computer with 4 cores i7-4790K CPU, GTX3090 graphic card, and 32GB memory was utilized for all coding activities.

4.4.1. Graph Neural Network

Fig. 4-5 and Fig. 4-6 demonstrate a successful training process for the GNN model. The close alignment between the training and validation loss curves suggests the model generalizes well to unseen data, minimizing the risk of overfitting. The smooth decline in loss, eventually reaching a plateau, signifies effective learning and convergence. Meanwhile, accuracy steadily improves, reaching around 93%. Together, these figures indicate that the training process achieved its goal: maximizing accuracy while minimizing loss.

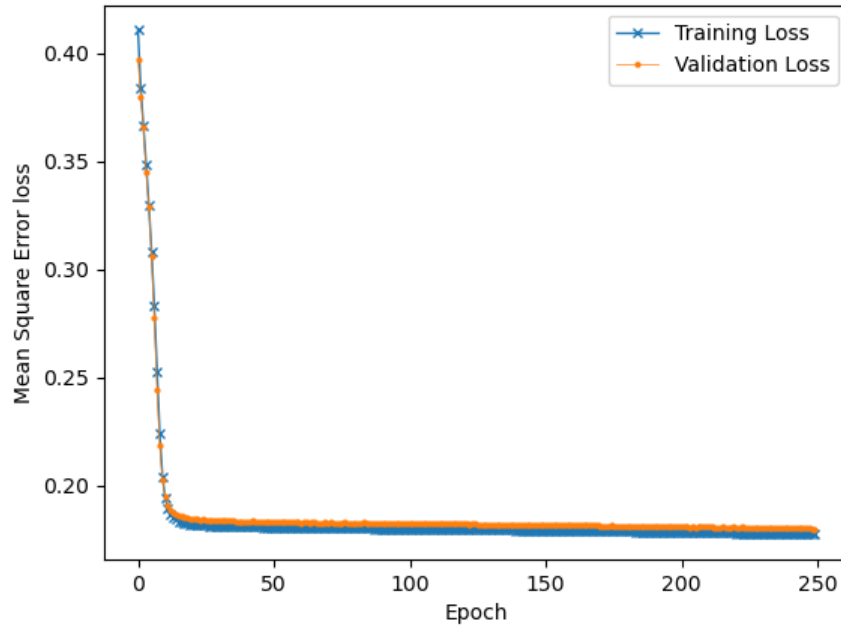


Fig. 4-5 MSE loss for training vs. validation of GNN in OFFLINE - training mode.

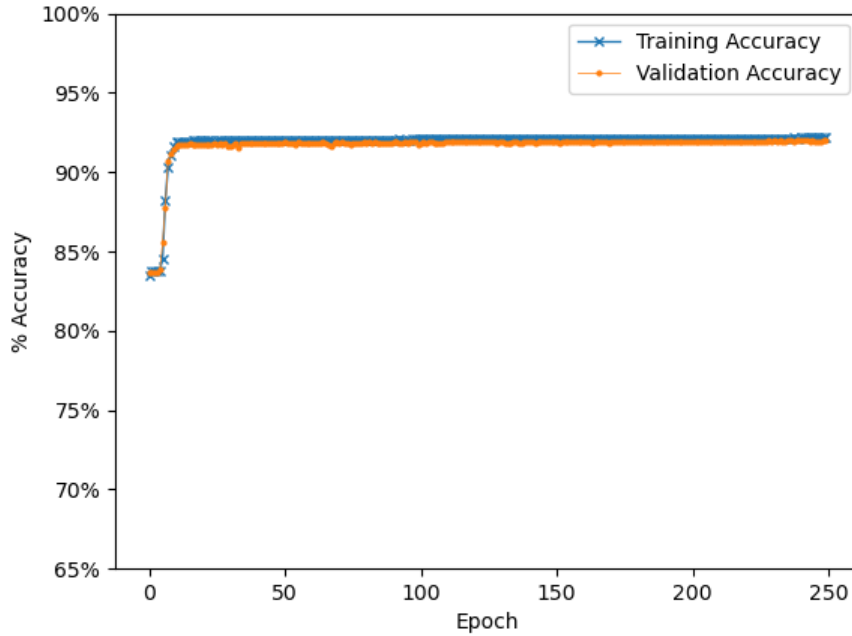


Fig. 4-6 Percent accuracy of prediction for training vs. validation of GNN model for the OFFLINE - training mode.

4.4.2. Performance Comparison on a Representative Sample

A representative sample was selected to illustrate the performance of the proposed RGNN-MILP method compared to other benchmark methods as detailed in TABLE 4-2, while statistical results over 1,000 samples will be provided in the subsequent subsections. The data values are expressed in absolute values as well as in percentages relative to the complete NR-OPF solution. The ideal method would provide a solution exactly matching the full NR-OPF solution, as close to 100% as possible with significantly reduced solving time.

TABLE 4-2 - Objective total cost and solving time of a representative sample.

	Total Cost		Solving Time	
	(%)	(\$)	(%)	(second)
Full NR-OPF	100	154,427.9	100	10.765
FGNN-LP	101.349	156,510.3	0.474	0.051
FGNN-MILP	99.998	154,424.9	84.742	9.122
RGNN-LP	110.079	169,992.7	0.520	0.056
RGNN-MILP	99.999	154,426.0	1.812	0.195

FGNN-LP and RGNN-LP demonstrate the fastest solving times compared to FGNN-MILP and RGNN-MILP. However, RGNN-LP shows the highest total cost among the methods evaluated, exceeding by up to 10%. FGNN-MILP has the longest solving time, approximately 84%. While FGNN-LP has the fastest solving time, it still has the second highest difference in total cost, over \$2,000 in absolute value. RGNN-MILP is the only method that achieves a total cost nearly equivalent to the full NR-OPF problem while reducing solving time by as much as 98%.

4.4.3. Total Cost Statistics

TABLE 4-3 offers substantial evidence that the FGNN-MILP and RGNN-MILP methods consistently produce solutions with objective total cost that are highly accurate, and closely aligned with the actual solution. The metric, total cost in percent, is defined in

$$\%Total\ Cost = \frac{Total\ Cost\ using\ GaNR - OPF\ methods}{Total\ Cost\ using\ Full\ NR - OPF}. \quad (4-11)$$

TABLE 4-3 - The total cost in percentage for the four methods using equation (4-11).

	Total Cost in Percentage (%)				
	Mean	Max	Min	Median	Std. Dev.
FGNN-LP	102.17%	103.72%	100.29%	102.23%	0.80%
FGNN-MILP	100.02%	100.98%	99.51%	100.00%	0.11%
RGNN-LP	102.25%	111.22%	100.00%	101.83%	1.75%
RGNN-MILP	100.05%	100.56%	99.92%	100.02%	0.09%

Both the median and the mean of these methods approach nearly 100%, indicating a reliable and precise performance. On the other hand, the FGNN-LP and RGNN-LP methods exhibit a much broader range of total costs, highlighting their variability in performance. The deviation from the actual solution in these methods can reach as high as 11%, indicating a less consistent and potentially less reliable approach. The significant difference in performance between the MILP-based methods and the LP-based methods underscores the superiority of the former in producing more accurate and stable results across different cases.

The observation is further underscored by Fig. 4-7, which visually depicts the clustering of the majority of solutions near the 100% total cost mark. Among the various methods, the RGNN-MILP method stands out for its exceptional performance. Notably, more than 80% of the solutions solved using this method perfectly align with the total costs of the associated full NR-OPF solutions, effectively matching the optimal solution. The RGNN-MILP method's ability to consistently achieve results that closely mirror

the actual cost of a full NR-OPF problem demonstrates its superior effectiveness in cost optimization.

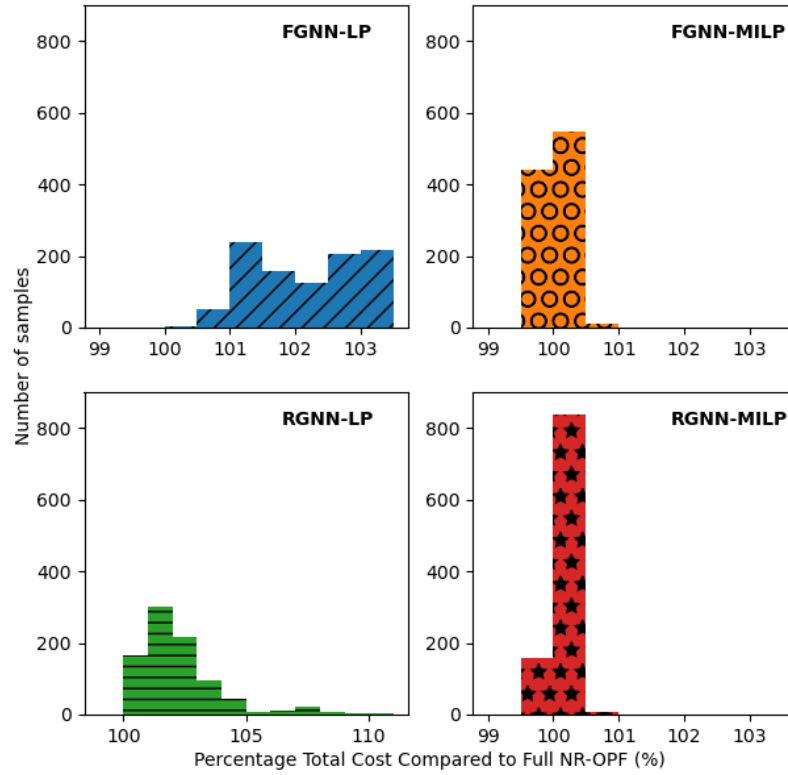


Fig. 4-7. Total cost of the four proposed benchmark methods.

4.4.4. Solving Time Statistic

TABLE 4-4 provides compelling evidence of remarkable reductions in solving time achieved for all four methods using the GaNR-OPF strategies. The FGNN-LP and RGNN-LP methods demonstrate particularly outstanding improvements, boasting mean solving times that are up to an astonishing 99% faster. The FGNN-MILP method also generally yields faster solving times, with an average reduction of around 40%. However, it is important to note that Fig. 4-8 reveals that certain samples processed by the FGNN-MILP method can take up to 10 times longer to solve compared to the

original full NR-OPF problem. This suggests that while the FGNN-MILP method is generally faster, it may encounter specific instances where it is less efficient.

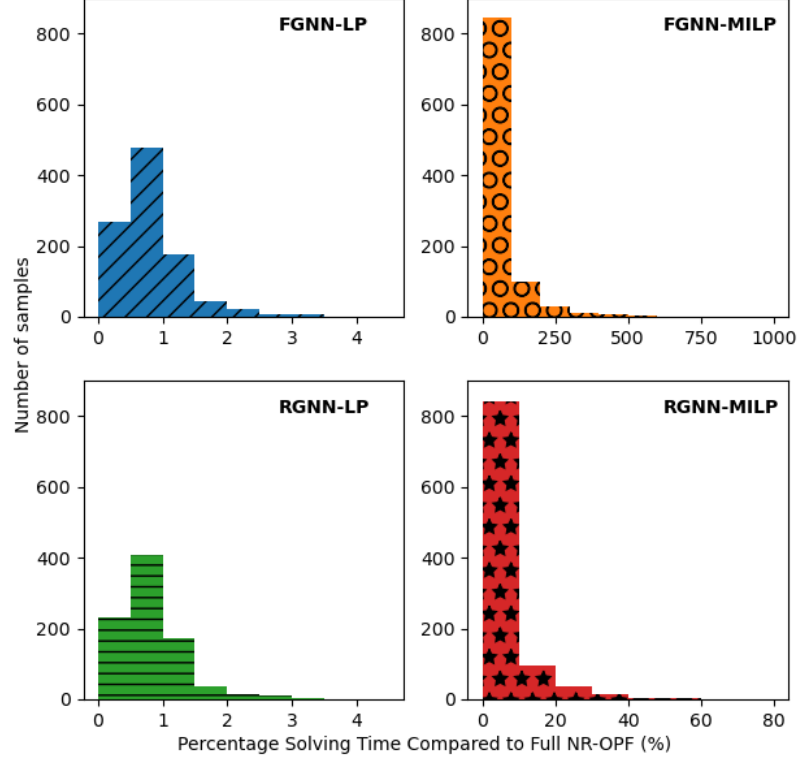


Fig. 4-8. Solving time of the four proposed benchmark methods.

In contrast, the proposed RGNN-MILP method, while not holding the absolute fastest solving time, still achieves impressive savings of 94% in solving times, with a median solving time of 3.5% for all samples. The consistent efficiency is highlighted, and the method underscores its potential as a valuable tool in reducing computational burden for NR-OPF problem.

$$\%Solving\ Time = \frac{Solving\ Time\ using\ GaNR - OPF\ methods}{Solving\ Time\ using\ Full\ NR - OPF} \quad (4-12)$$

TABLE 4-4 - Solving time in percentage for the four methods using equation (4-12).

	Solving Time in Percentage (%)				
	Mean	Max	Min	Median	Std. Dev.
FGNN-LP	0.81%	3.38%	0.08%	0.72%	0.49%
FGNN-MILP	60.23%	1085.76%	2.72%	30.09%	92.89%
RGNN-LP	0.83%	4.01%	0.08%	0.73%	0.51%
RGNN-MILP	6.34%	83.77%	0.41%	3.47%	8.24%

4.4.5. RGNN-MILP Method

Among the four benchmark methods, the proposed RGNN-MILP method emerges as a clear frontrunner for GaNR-OPF model in terms of overall speed and consistency. Its success can be attributed to the powerful combination of pre-ML and post-ML steps, which work in tandem to dramatically reduce the complexity of MILP problems. The reduction in complexity translates to an impressive 80% decrease in the number of critical lines required for the final solving NR-OPF step, streamlining the computational process significantly.

Remarkably, the RGNN-MILP method not only simplifies the problem but also maintains the quality of the solutions. It achieves solutions with a total cost that is virtually identical to the full NR-OPF solution, all while delivering a substantial 94% reduction in solving time on average. This exceptional performance highlights the RGNN-MILP method's unique ability to generate high-quality solutions without compromising computational efficiency.

The significant speedup is made possible by leveraging the predictive power of the GNN model and the synergistic interplay between the pre- and post-ML selection steps. By intelligently reducing the problem's complexity without sacrificing solution accuracy, the RGNN-MILP method offers a promising avenue for tackling computationally intensive tasks in a more efficient and effective manner.

4.5. Conclusion

The proposed RGNN-MILP method demonstrates superior performance in addressing network reconfiguration challenges. Using this method for GaNR-OPF achieves near-identical objective total costs compared to the full NR-OPF problem while drastically reducing computational time by approximately 94%. While other alternative methods may offer marginally faster solutions, they often do so at the expense of overall solution accuracy and optimality. As for the proposed RGNN-MILP method for GaNR-OPF, the integration of pre-ML filtering and post-ML selection mechanisms proves to be crucial in achieving this optimal balance between solution speed and quality. GaNR-OPF has proven highly effective in managing power system network reconfiguration problems and determining the optimal network topology. Although our current research is centered on a moderate-scale IEEE 73-bus system, extrapolation suggests comparable computational resource savings are achievable when scaling to larger systems comprising thousands of buses.

Further research will delve into alternative network reconfiguration strategies, including imposing constraints for line switching operations and utilizing virtual buses with distinct topologies for granular modeling of loads and generators. Additionally, the

integration of GNNs into broader advanced OPF problem domains presents a promising avenue for future exploration.

5. CONSTRAINTS AND VARIABLES REDUCTION FOR OPTIMAL POWER FLOW USING HIERARCHICAL GRAPH NEURAL NETWORKS WITH VIRTUAL NODE-SPLITTING

Power system networks are often modeled as homogeneous graphs, which limits the ability of graph neural network (GNN) to capture individual generator features at the same nodes. By introducing the proposed virtual node-splitting strategy, generator-level attributes like costs, limits, and ramp rates can be fully captured by GNN models, improving GNN's learning capacity and prediction accuracy. Optimal power flow (OPF) problem is used for real-time grid operations. Limited timeframe motivates studies to create size-reduced OPF (ROPF) models to relieve computational complexity. In this paper, with virtual node-splitting, a novel two-stage adaptive hierarchical GNN is developed to (i) predict critical lines that would be congested, and then (ii) predict base generators that would operate at the maximum capacity. This will substantially reduce the constraints and variables needed for OPF, creating the proposed ROPFLG model with reduced monitor lines and reduced generator-specific variables and constraints. Two ROPF models, ROPFL and ROPFG, with just reduced lines or generators respectively, are also implemented as additional benchmark models. Case studies show that the proposed ROPFLG consistently outperforms the benchmark full OPF (FOPF) and the other two ROPF methods, achieving significant computational time savings while reliably finding optimal solutions.

5.1. Literature Review

Optimal Power Flow (OPF) is an essential tool in power system operations, analysis, and scheduling, including real-time economic dispatch. It involves optimizing an objective function, such as minimizing generation, while adhering to a set of physical and operational constraints, like generation limits and line thermal ratings. The scope of OPF applications has significantly expanded due to its ability to address various challenges within the power network, from grid management to resource allocation.

With the rapid growth of renewable energy integration, particularly from sources like wind and solar, the intermittent nature of these resources introduces new complexities to the system. Generation capacity can fluctuate dramatically throughout the day, making it crucial for OPF calculations to be both fast and highly accurate to ensure demand is met without compromising the reliability of the grid.

To meet these evolving challenges, existing OPF models must be improved. This includes refining algorithms to handle the uncertainty and variability of renewable energy, enhancing computational efficiency for real-time application, and ensuring scalability as power systems grow more complex. Advances in OPF are vital to supporting the transition to cleaner energy while maintaining the reliable operation of the power grid.

Machine learning (ML) has been widely applied to address power flow problems [9]. Graph neural network (GNN) is an advanced neural network architecture tailored for graph-structured data, leveraging message passing between nodes to capture complex dependencies. The topology of the network is essential in GNNs, as it allows information to flow between adjacent nodes and edges through multiple layers,

capturing both local and global relationships within the graph [15], [16]. GNNs have been implemented in various types of OPF problems [30], [28]. For instance, in [58], GNNs are used to predict $N-1$ contingencies, reducing computational time for solution discovery. Additionally, GNNs have been applied to network reconfigured OPF problems, where they optimize network topology to enhance solution efficiency [62].

Modifying the topology of a transmission network through techniques, such as line switching and bus splitting, has been previously employed as a corrective mechanism to mitigate congestion issues [63], [64]. In the context of transmission expansion planning, bus splitting has demonstrated its efficacy in enhancing optimal dispatch solutions [65]. The process involves physically altering the system topology by subdividing a single node into multiple nodes [66].

Network topologies of power systems are commonly represented as homogeneous graphs, where all nodes and edges are of a single type. Homogeneous graph restricts the GNN model from capturing distinct node features when multiple generators are associated with a single node. By subdividing virtual nodes from existing physical nodes, additional generator-specific features can be incorporated into the graph network. Since GNNs are highly sensitive to network topology, virtual node-splitting serves as an effective technique for enhancing network structure. Using virtual node-splitting, attributes for each individual generator such as generation costs, generation limits, and ramping rates can be utilized to extract critical information, thereby improving learning capacity and efficiency of the GNN model.

In our prior work, we developed an ML algorithm utilizing GNNs to improve the formulation of a reduced optimal power flow (ROPF) problem [35]. The GNN

model was tailored to predict congested transmission lines under varying load profiles, enabling the identification of a critical subset of lines for active monitoring. By concentrating only on these critical lines, the approach reduces the number of line flow constraints within the OPF problem, effectively simplifying the problem and enhancing the efficiency of the ROPF formulation. The method not only accelerates OPF computations but also supports more scalable solutions, which are especially beneficial for systems with high variabilities in load profiles and generation capabilities, where real-time performance is essential.

In this paper, we present further advancements in the formulation of the ROPF problem, with key contributions and innovations summarized as follows:

- In contrast to the previous GNN model, which combined the capacities of all generators at each node as a single feature, this research employs the proposed virtual node-splitting strategy that enables GNN to explicitly model each generator individually. The approach substantially increases the number of node's features available for learning, thereby improving the accuracy of GNN predictions.
- An adaptive hierarchical GNN model is developed to predict generators that would operate at their maximum capacities, effectively reducing the number of generation variables and constraints and expediting the OPF solution process.
- Three ROPF methods—(i) ROPFL with reduced monitoring lines that are predicted to be congested, (ii) ROPFG with reduced variables and constraints related to the generators that are predicted to operate at their

maximum power ratings, and (iii) ROPFLG with both reduced monitor lines and reduced generator-specific variables and constraints—were developed and benchmarked against the baseline full OPF (FOPF). Case studies demonstrate that the proposed ROPFLG consistently outperforms the benchmark FOPF as well as the other two ROPF methods. Leveraging the hierarchical GNN model, the proposed ROPFLG method achieves substantial reductions in computation time while consistently identifying the optimal solution.

5.2. Backgrounds

5.2.1. Reduced Optimal Power Flow using Lines and Generators

The OPF model is commonly used for real-time economic dispatch in power systems, optimizing generation dispatch to meet system loads while satisfying various constraints within the transmission network. Typically, the OPF objective function aims to minimize operating costs. In the OPF problem below, the total generation cost is minimized as expressed in

$$\mathbf{Obj} : \min \sum_{g \in G} c_g P_g \quad g \in G \quad (5-1)$$

with decision variables representing generation P_g and the corresponding cost c_g .

$$P_g^{\min} \leq P_g \leq P_g^{\max} \quad g \in G \quad (5-2)$$

establishes the minimum and maximum generation limits for each generator within the system. The power flow P_k on each transmission line is calculated using

$$P_k = \frac{\theta_{f(k)} - \theta_{t(k)}}{x_k} . \quad k \in K \quad (5-3)$$

In constraint (5-3), $\theta_{f(k)}$ and $\theta_{t(k)}$ represent the phase angles at the sending and receiving ends of line k . The line limit constraint in

$$-RateA_k \leq P_k \leq RateA_k \quad k \in K \quad (5-4)$$

restricts P_k to be within the line's thermal rating limit, $RateA_k$.

The constraint below

$$\sum_{g \in G(n)} P_g + \sum_{k \in K(n-)} P_k - \sum_{k \in K(n+)} P_k = d_n \quad n \in N \quad (5-5)$$

enforces nodal power balance, ensuring that the power generation meets the given nodal load d_n at each bus.

In most OPF problems, the optimal solution is typically limited by a number of congested transmission lines. Independent System Operators (ISOs) often monitor a specific subset of lines that experience high loading or congestion. These subsets of lines are selected based on historical OPF solutions and load profiles. Similarly, maximum generation limits impose constraints on certain generators. Identifying generators that consistently operate at maximum capacity allows for a reduction in the total number of constraints.

Our previous work focused on dynamically selecting these critical lines in response to specific load profiles, utilizing machine learning, particularly GNN. In this study, we extend this approach by proposing a GNN model to predict which generators will reach maximum capacity under varying load conditions.

The hierarchical GNN model is designed to accurately forecast both congested

lines and generators at their maximum generation for dynamic load profiles. By applying GNN predictions for line flows and generation levels, we can significantly reduce the number of constraints. The approach requires monitoring only a smaller subset of critical lines and dispatching a selected number of online generators within the OPF problem, resulting in a ROPF problem that alleviates real-time computational demands. Based on the predictions from the hierarchical GNN model, constraints for non-congested lines are ignored and parameters for maximum capacity generators are fixed during the formulation stage of ROPF problems. Ultimately, instead of applying constraints (5-2) and (5-4) across all lines and generators, we implement constraints below:

$$P_y^{min} \leq P_y \leq P_y^{max} \quad y \in Y \quad (5-6)$$

$$-RateA_x \leq P_x \leq RateA_x \quad x \in X \quad (5-7)$$

for a subset of congested lines X and generators with undetermined generation levels Y .

By incorporating predictions from the GNN model, the ROPF solution identifies the optimal combination of generation outputs to satisfy the given load at minimal cost, while adhering to line limits and maintaining nodal power balance constraints. The GNN model can be updated in near-real-time with varying load profiles across different scenarios, enabling system operators to achieve optimal generation dispatch within the allocated computing time.

5.2.2. Virtual Node-Splitting

Machine learning algorithms have been widely applied in power system optimization [8], [27]. However, these models often overlook the inherent topological

information within electrical networks that could significantly improve their learning efficiency. GNN, a specialized class of ML models, address this by optimizing transformations across all graph components—nodes, edges, and global context—while maintaining graph symmetries, permutation invariance.

The unique capabilities of graph neural networks serve as powerful tools for advancing power system optimization by capturing and modeling the complex interdependencies within networked systems. Unlike other ML methods, GNNs leverage the structural properties of the network, enabling them to capture both local and global relationships across nodes and edges, which is essential for accurately predicting critical variables such as line flows and generation levels. The ability to integrate network topology into the learning process offers substantial improvements in prediction accuracy and computational efficiency.

In GNNs, network topology is incorporated through an adjacency matrix, an nb by nb matrix, where nb represents the number of buses. The adjacency matrix facilitates a more comprehensive representation of node and edge features. In most electrical networks, topology encodes essential, often implicit features that can be learned and extracted to enhance the model's performance.

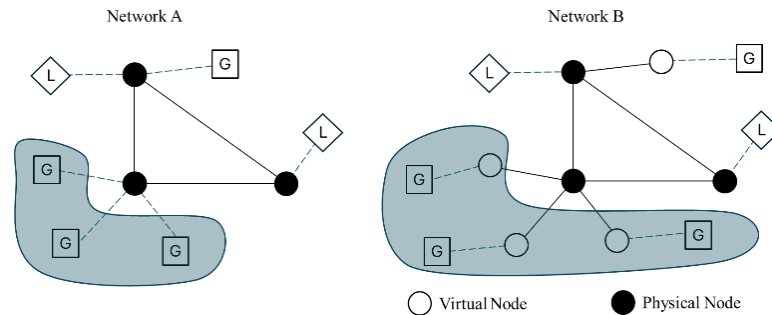


Fig. 5-1. A 3-bus system (Network A) vs. the expanded 3-bus system using virtual node-splitting (Network B).

As shown in Fig. 5-1, a simple 3-bus power system in Network A was augmented using virtual node-splitting and transformed into Network B. In our previous work with a homogeneous graph structure, the number of features per node or edge was restricted, preventing individual generator modeling at the node level. Generators at each bus were (e.g., the three generators in the shaded region) aggregated into a single equivalent generator, representing total generation at that node. Consequently, node-level generator-specific features, such as generation cost and ramp rate, were discarded. By using a virtual node-splitting, each generator can be represented as connected to a virtual node (white circle), while loads remain in the real nodes (black circle), allowing all generator-specific features to be used in GNN training.

GNNs leverage message passing, made possible by the adjacency matrix. Introducing virtual nodes increases the total number of nodes to $(nb + ng)$, where ng is the number of generators. The adjacency matrix is expanded into size of $(nb + ng)$ by $(nb + ng)$. Although a larger GNN model with an expanded adjacency matrix may increase training time, it is expected to enhance learning performance. Once the GNN model is trained, predictions are made quickly. In GNNs, the more features provided per node, the more effectively the model can learn the network's global context, establishing relationships among various node and edge features.

By providing faster and more precise solutions, GNNs are particularly valuable for tackling the complex optimization challenges faced in modern power systems, such as those posed by high penetration of renewable energy sources and dynamic load conditions. As these networks evolve in scale and complexity, GNNs offer a complementary enhancement to traditional methods, using the inherent graph structure

to gain deeper insights, streamline computation, and improve the robustness of optimization outcomes. Utilizing virtual nodes to expanded network topology for training GNN models holds promise for significantly improving operational efficiency and scalability in real-time applications within the power industry.

5.3. Methodologies

The GNN models and OPF algorithms were implemented in Python, utilizing the pyomo library and the gurobi solver for solving OPF problems [21]. The GNN models were developed with the spektral library [23], while figures and graphs were generated using matplotlib library. All computational tasks were performed on a system with a 4-core Intel i7-4790K CPU, a GTX 3090 GPU, and 32 GB of memory. To create a robust dataset for training and testing the GNN model, we conducted full OPF simulations on the IEEE 73-bus system under varied load profiles, generating 10,000 samples. For each sample, the load profile was adjusted within $\pm 10\%$ of the base load profile to ensure sufficient variability.

5.3.1. OFFLINE – Training Stage

The generated dataset, consisting of 10,000 samples, is partitioned into training and validation subsets. Labels for each transmission line are assigned based on a predetermined loading threshold, defined as a percentage of the line's thermal rating. For instance, if the thermal rating of a line is 200 MW and the loading threshold is set at 70%, the line is classified as congested when the power flow exceeds 140 MW. From Fig. 5-2, the first GNN model was trained to predict congested lines. Predictions for congested lines from the first GNN model will then be used as features to train the

second GNN model. The second GNN model is then used to predict which generator will produce at maximum capacity. By knowing which lines will congest and which generators will produce at maximum capacity, the number of constraints in the FOPF problem can be significantly reduced resulting in a ROPF problem. The ROPF problem is then solved, and the solution is verified that all constraints have been met.

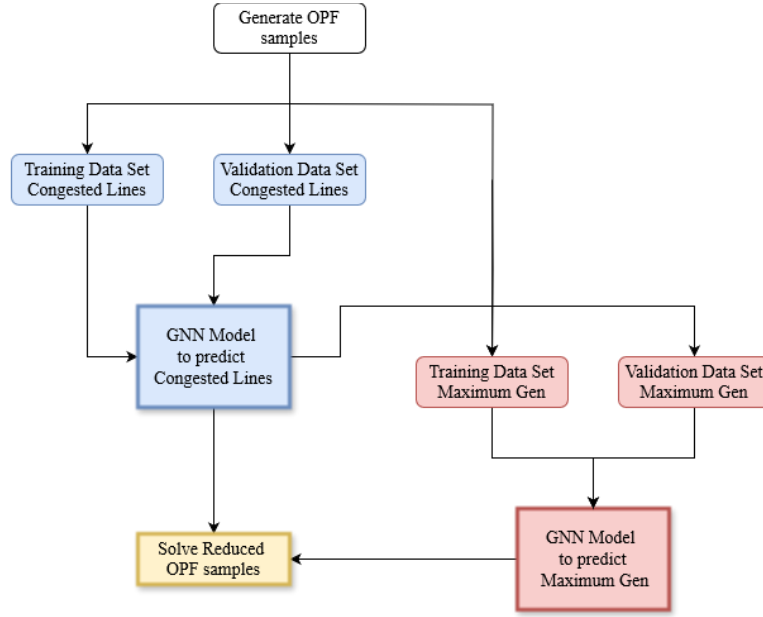


Fig. 5-2. Flow chart of proposed hierarchical GNN using virtual node-splitting.

5.3.2. ONLINE – Operational Stage

In the operational stage, the trained GNN models were deployed to process real-time test samples, predicting both congested lines and maximum-capacity generators. Four methods were implemented:

- FOPF method: solves the OPF problem without reducing the number of constraints, monitoring all lines and dispatching all online generators, which serves as a baseline.

- **ROPFL method:** the GNN model predicts congested lines. Line limit constraints for non-congested lines are removed, resulting in a ROPF problem.
- **ROPFG method:** the GNN model predicts maximum-capacity generators. Parameters for maximum-capacity generators are set, resulting in a different ROPF problem.
- **Proposed ROPFLG method:** the hierarchical approach uses the GNN model to first predict congested lines. These line predictions are then included as input features for the second GNN model, which predicts the generators that produce at their maximum capacity. Line limit constraints for non-congested lines are removed, and variables for predicted maximum-capacity generators are switched to known parameters that would reduce both variables and constraints, resulting in a further smaller ROPF problem.

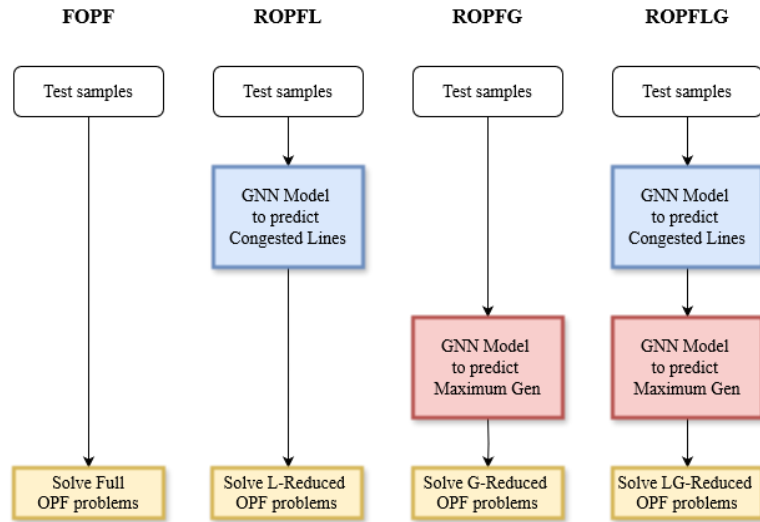


Fig. 5-3. Flow chart of three proposed ROPF and FOPF.

The flowchart in Fig. 5-3 summarizes the four proposed methods. Once the ROPF problem is formulated, it is solved, and its solution is verified for validity and optimality. By narrowing the focus to a smaller subset of critical constraints, the ROPF problem significantly decreases computational complexity, potentially enabling faster and more efficient solutions. The solving time of each ROPF method is then compared to that of the baseline FOPF method to assess performance improvements.

5.4. Results and Analysis

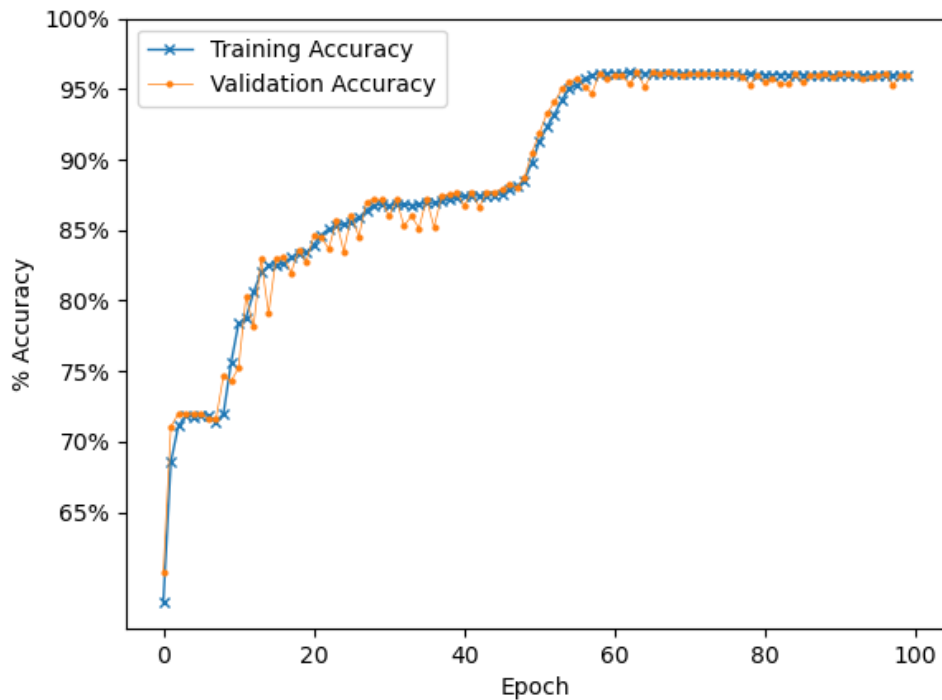


Fig. 5-4. Percent accuracy of prediction for training vs. validation of the GNN - Maximum Gen model.

This section evaluates the hierarchical GNN model's performance during training. As shown in Fig. 5-4,, the model demonstrates strong performance over 100 training epochs, with training accuracy closely tracking validation accuracy. Both

figures confirm efficient training. The close alignment of training and validation loss curves in Fig. 5-5 suggests effective generalization to unseen data, minimizing overfitting risk. The smooth, consistent decrease in loss, eventually plateauing, indicates stable learning and convergence. Meanwhile, accuracy steadily rises during training, reaching approximately 96%. Overall, these results demonstrate the training process can effectively balance high accuracy with minimized loss.

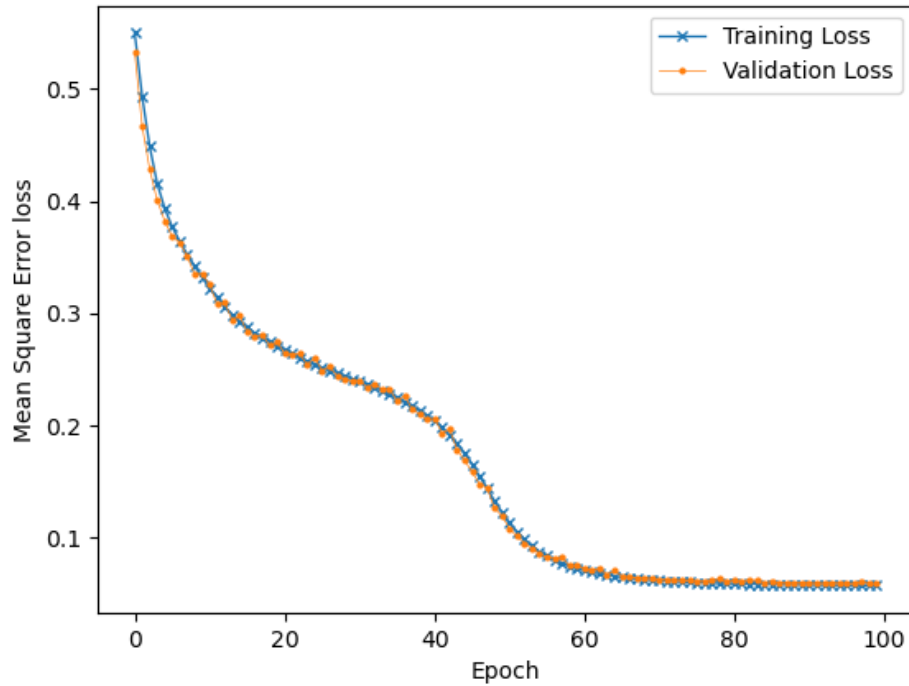


Fig. 5-5. MSE loss for training vs. validation of the GNN - Maximum Gen model.

The trained GNN models were evaluated on a testing dataset of 1,000 samples, generating predictions for each sample regarding congested lines and maximum-capacity generators. These predictions present two potential error types:

- Type I error (false positive): Incorrectly classifying a non-congested line as congested or a generator as operating at maximum capacity when it is not.

- Type II error (false negative): Failing to identify an actual congested line or a generator operating at maximum capacity.

Previous work on ROPF indicate that solutions with a high Type II error rate (false negatives) are more likely to violate system constraints. There is a strong correlation between missing predictions for congested lines and maximum-capacity generator that can lead to invalid solutions for ROPF problem.

TABLE 5-1 - Percent error of prediction for congested lines and maximum generators.

	False Positive	False Negative	Total Error
Congested Lines	1.07%	0.12%	1.19%
Maximum Generator	5.35%	0.92%	6.27%

Since predictions for congested lines serve as features for predicting maximum-capacity generators, a small error rate in congested line predictions may slightly increase the error rate for generator capacity predictions. As shown in TABLE 5-1, the overall error rate for predicting maximum-capacity generators is slightly elevated at over 6%, but the false negative rate remains low, under 1%. False negative errors for predictions of congested line are excellent, with an error rate of approximately 0.1%. With these low false negative rates for both congested lines and maximum-capacity generators, the likelihood of generating invalid ROPF solutions from these predictions is expected to be minimal.

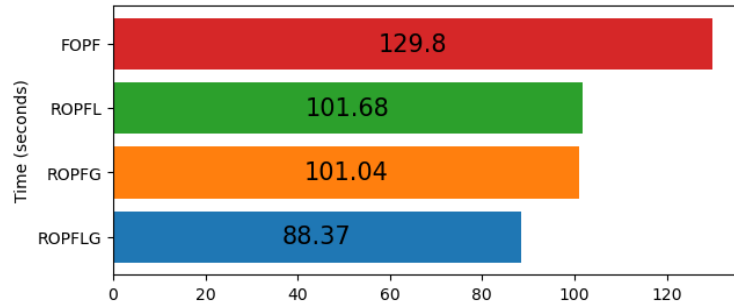


Fig. 5-6. Solving time using the four methods for 1,000 test samples.

In Fig. 5-6 and TABLE 5-2, it compares the computation times for solving 1,000 OPF problems using the four methods. The FOPF method serves as the baseline, taking 129.80 seconds, which is set at 100%. The ROPFL method reduces the computation time to 101.68 seconds, achieving almost 22% of saving in solving time. Similarly, the ROPFG method takes 101.04 seconds, reaching a similar percentage of time saved. The most efficient method, ROPFLG, completes in 88.37 seconds, which is 32% faster than FOPF. The average total cost for 1000 samples is also shown in TABLE 5-2. For the three ROPF methods, the mean total cost is almost identical to the benchmark FOPF model, with ROPFL and ROPFG slightly higher than ROPFLG; the difference is well below 0.1% which is negligible.

TABLE 5-2 - Mean total cost and time save in (%) for solving 1,000 test samples.

	Mean Total Cost (%)	Time Saving (%)
FOPF	100%	0
ROPFL	100.061%	21.67%
ROPFG	100.064%	22.16%
ROPFLG	100%	31.92%

The computational savings achieved with the hierarchical ROPFLG method, though substantial, are somewhat less than anticipated. It can be attributed to the limited number of constraints that can be removed before rendering the OPF problem unsolvable. Additionally, the computational benefits of ROPFLG are expected to be more pronounced in larger systems featuring multiple congested lines and several generators operating at maximum capacity. Overall, ROPFLG achieves the most significant reduction in computation time compared to other methods while maintaining the same optimal solution as the baseline FOPF.

5.5. Conclusion

Among the four methods assessed in this study, the ROPFLG approach demonstrates the greatest computational time savings in finding solutions for OPF problems. All solutions obtained using the ROPFLG method are optimal, with no violations of operational constraints. Furthermore, the objective cost for each solution aligns closely with that of the baseline FOPF. Although the chosen ROPFLG method is more complex than FOPF—requiring the training of a hierarchical GNN model—the efficiency gains in solution speed are significant during the operational stage, especially for large-scale OPF problems.

The study also opens pathways for further exploration of GNN model variations. For instance, a transfer learning GNN model could be investigated, where the congested line GNN model is leveraged to predict maximum-capacity generators without additional retraining. Alternatively, a dual prediction GNN model could be developed to concurrently predict congested lines and maximum-capacity generators.

Additionally, virtual node-splitting could be applied to address various OPF problem variants, such as N - contingency OPF or network-reconfigured OPF. Virtual node-splitting allows for the inclusion of more node-level features, thereby enhancing the model's capacity to learn and extract critical information, which ultimately improves prediction performance.

6. CONCLUSION AND FUTURE WORK

The United States is experiencing an unprecedented growth in both energy demand and generation in the last few years, driving a significant transformation of the electrical grid. Unlike the traditional unidirectional power flow paradigm, modern grids are increasingly characterized by bidirectional power flow. Consequently, network operators must adopt advanced methodologies to swiftly detect instability while minimizing system costs. Enhancing traditional OPF approaches with ML, particularly GNN, has emerged as a viable solution to address these challenges.

To ensure a sustainable and reliable grid, strategic long-term investments are essential. Strengthening, securing, and continuously upgrading the national electrical grid must remain a top priority. Adequate funding should be allocated to research initiatives focused on overcoming existing and emerging challenges associated with integrating high levels of renewable energy. Addressing these issues requires a coordinated effort from all stakeholders, fostering a collective commitment to achieving a greener, more sustainable future. By proactively tackling these challenges, we can establish an environmentally resilient and future-ready grid for generations to come.

6.1. Contributions

The ROPF method has been demonstrated as an effective approach for solving OPF problems efficiently while maintaining high solution quality. Compared to NN and CNN, the error prediction results indicate that GNN is the most effective model for classifying branches likely to become overloaded or congested under a given load profile. GNN achieves the lowest error prediction rate across all thresholds, which

directly translates to reduced computation time in the proposed ROPF model. The selection of an appropriate loading threshold for branch classification is also critical, as accuracy must be balanced with computational efficiency. At an 85% threshold, the GNN model predicts only 0.4% of solved ROPF cases with at least one branch violating its rating limit while achieving a 16% reduction in computation time, a notable improvement in resource efficiency and computational cost.

In Chapter 2, model selection emerges as a key step in training ML algorithms. The GNN model, which retains node features, edge features, and network topology during training, demonstrates superior performance compared to NN and CNN. While there are challenges associated with building and training GNN models, such as high data requirements and computational expense, the benefits are significant. Once trained, the GNN model can deliver rapid and accurate predictions.

Our work in Chapter 3 has successfully minimized the number of constraints in the comprehensive $N-1$ ROPF problem through the development of an AHGNN model. By leveraging this model, we formulated a streamlined $N-1$ ROPF algorithm that integrates predictions from three distinct sources. This strategic fusion of predictions achieved a notable 19% reduction in solving time while maintaining the integrity and quality of the solutions. The adoption of this diverse and strategic approach underscores the capability of the AHGNN model to significantly enhance computational efficiency in addressing $N-1$ ROPF problems.

The proposed RGNN-MILP method exhibits exceptional performance in addressing network reconfiguration challenges. When applied to GaNR-OPF, it achieves near-identical total objective costs compared to the full NR-OPF problem,

while significantly reducing computational time by approximately 94%. Although alternative methods may offer slightly faster solutions, they often compromise solution accuracy and optimality. The RGNN-MILP method's integration of pre-ML filtering and post-ML selection mechanisms is pivotal in achieving an optimal balance between computational efficiency and solution quality. In Chapter 4, GaNR-OPF has proven to be a highly effective tool for managing power system network reconfiguration challenges and determining optimal network topologies.

In the previous chapter, the ROPFLG approach achieves the highest computational time savings in solving OPF problems among the four methods evaluated in this study. All solutions derived using the ROPFLG method are optimal, with no operational constraint violations, and the objective costs closely match those of the baseline FOPF method. While the ROPFLG method introduces additional complexity due to the need for training a hierarchical GNN model, the substantial efficiency gains during the operational stage make it particularly advantageous.

6.2. Future Work

From our learned experience in developing and training GNN models, we are confident that applications of GNN for more complex and practical OPF models can achieve even greater computational time savings while delivering superior results. Our work focused only on the IEEE 73-bus system as a modest benchmark. However, future work can extend toward scaling to larger systems with thousands of buses. The adaptability of GNN models ensures that accurate predictions can be maintained across various power system network topologies.

The hierarchical GNN model is a distinctive architecture designed to enhance the predictive capabilities of GNNs. However, two alternative ML architectures could also be explored to further optimize the design and performance of GNN models:

- **Transfer Learning in GNNs:** This approach involves leveraging a pre-trained GNN model, such as one designed for congested-line prediction, to predict maximum-capacity generators without the need for additional retraining. Transfer learning can enable faster deployment and reduce the computational cost associated with training new models from scratch. By repurposing learned features, this method could improve the adaptability and efficiency of GNN applications in OPF-related tasks.
- **Dual Prediction GNNs:** This architecture aims to develop a GNN model capable of simultaneously predicting multiple outputs, such as congested lines and maximum-capacity generators. A dual prediction approach can optimize resource utilization by addressing multiple objectives within a single model, potentially improving prediction accuracy for both outputs by capturing shared features and interactions across the tasks while decreasing training time.

Both architectures have demonstrated effectiveness in various fields of research and could offer significant benefits when applied to power systems. These approaches warrant exploration to evaluate their impact on predictive performance, computational efficiency, and scalability compared to the hierarchical GNN model. A comparative analysis of these architectures would provide valuable insights into their strengths and

trade-offs, enabling the selection of the most suitable design for specific OPF problem scenarios.

Virtual node-splitting has emerged as an innovative technique to enhance the predictive performance of GNN models by increasing the number of node-level features available for learning. The technique facilitates the exchange of additional information across nodes and edges during the training phase, leading to a significant improvement in the model's learning capacity and prediction accuracy.

Our research work has successfully applied virtual node-splitting to traditional OPF problems, demonstrating its effectiveness in capturing complex system dynamics. Given its proven advantages, virtual node-splitting is a promising candidate for application to other OPF problem variants, such as $N-1$ OPF and NR-OPF. For $N-1$ OPF, virtual node-splitting could enhance the model's ability to predict post-contingency violations and identify optimal corrective actions. Similarly, in NR-OPF, the technique could improve the accuracy of topology reconfiguration decisions by incorporating more granular node-level features, such as localized load and generation attributes.

Adopting virtual node-splitting for these advanced OPF variants holds significant potential for advancing predictive accuracy, scalability, and solution efficiency in power system optimization.

In this dissertation, GNNs have been employed to address various variants of the OPF problem, with a focus on the Direct Current OPF (DC-OPF) formulation. While the current implementation focuses on DC-OPF, extending GNN methodologies to Alternating Current OPF (AC-OPF) is a logical progression. Given the observed

performance of GNNs for DC-OPF applications, their application to AC-OPF is expected to be feasible with appropriate modifications.

The formulation of the OPF problem in this dissertation derives the line flow equation based on the angular difference between connected buses and the reactance of the transmission line. However, industry standards typically utilize line shift-factors to calculate the changes in line flows. Since shift-factors inherently depend on the impedances of all network paths, their use enhances the practicality of the OPF formulation and aligns the problem more closely with standard industry practices.

The application of GNN in power systems remains an emerging field, but their potential to drive advancements in power system technologies is significant. The proposed GNN model demonstrates substantial improvements in computational efficiency for solving OPF problems and its variants, highlighting its value as an effective tool for optimizing grid operations. These enhancements not only improve computational speed but also open the door for GNNs to address a wider range of challenges within power systems.

The continued development and application of GNNs in power systems are poised to broaden their role in optimizing system performance, improving reliability, and enhancing decision-making processes. As machine learning technologies continue to evolve, GNNs are expected to play a central role in advancing the field of power system optimization, demonstrating the transformative impact of these technologies in the evolving landscape of smart grids and energy systems.

6.3. Summary of Publications

- Thuan Pham and Xingpeng Li, “Neural Network-based Power Flow Model,” in IEEE Green Technologies Conference, Houston, 2022 [9].
- Thuan Pham and Xingpeng Li, “Reduced Optimal Power Flow Using Graph Neural Network,” in 54th North American Power Symposium, Salt Lake City, 2022 [35].
- Thuan Pham and Xingpeng Li, “N-1 Reduced Optimal Power Flow Using Augmented Hierarchical Graph Neural Network,” In review. Available on arXiv as a preprint. <https://arxiv.org/abs/2402.06226> [58].
- Thuan Pham and Xingpeng Li, “Graph Neural Network-Accelerated Network-Reconfigured Optimal Power Flow,” In review. Available on arXiv as a preprint. <https://arxiv.org/abs/2410.17460>. [62]
- Thuan Pham and Xingpeng Li, “Constraints and Variables Reduction for Optimal Power Flow Using Hierarchical Graph Neural Networks with Virtual Node-Splitting,” In review. Available on arXiv as a preprint. <https://arxiv.org/abs/2411.06268> [67].

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